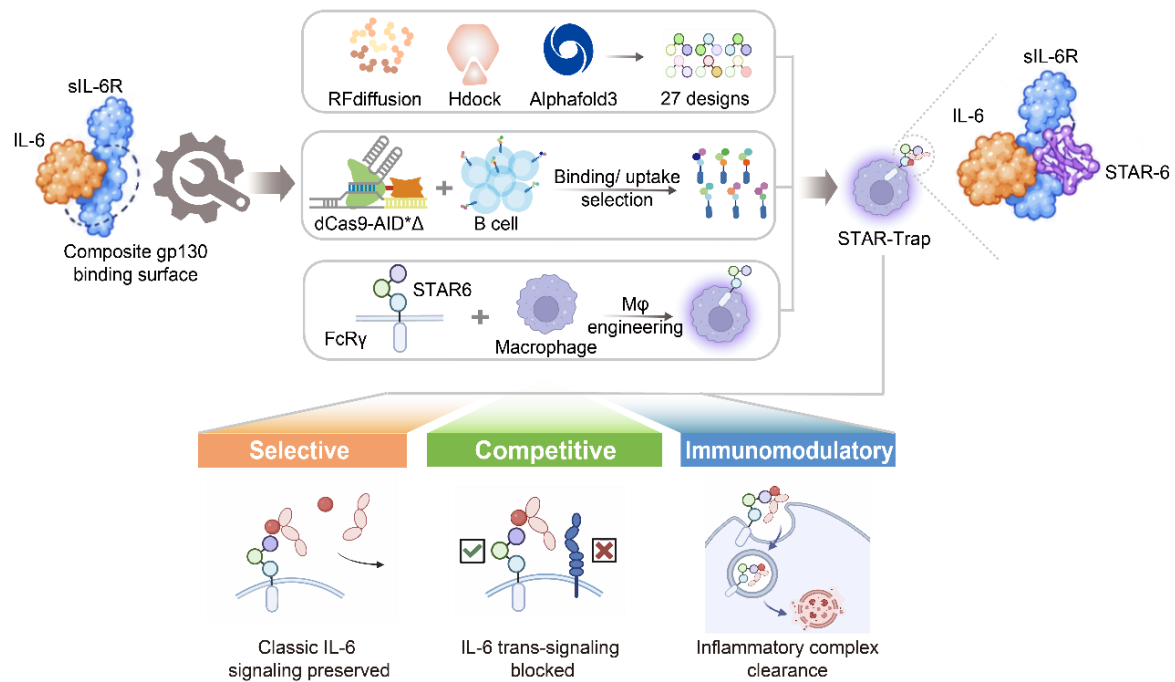
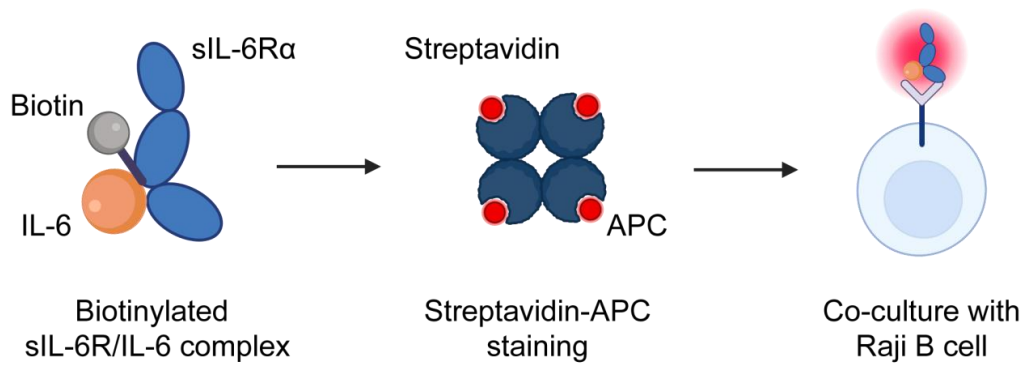




Extended Fig 1. Computational design and directed evolution of complex-specific binders.

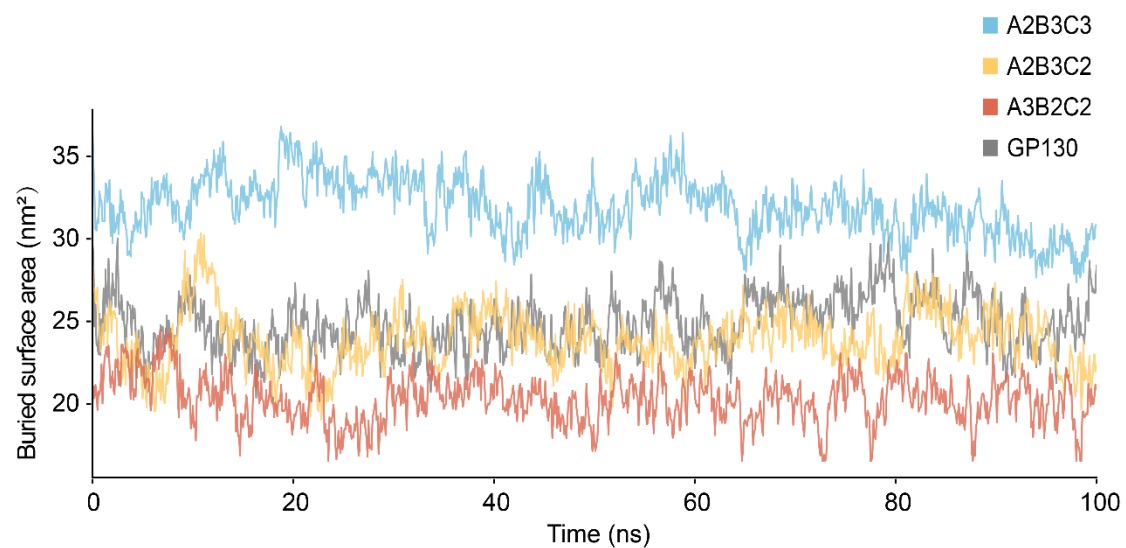
The composite gp130-binding surface formed by the IL-6/sIL-6R complex was used as the target template. Candidate binders were generated via RFdiffusion, Hdock, and AlphaFold3, yielding 27 initial designs. These candidates were displayed on B cells and subjected to binding and uptake selection. A dCas9-AID*Δ system was used to introduce targeted mutations in the candidate genes, followed by functional screening to enrich for high-affinity variants (STAR-6). The optimized binder was then fused with FcRγ and engineered into macrophages to generate STAR-Trap-expressing macrophages.

Functional properties of STAR-Trap. The engineered macrophages exhibit three key properties. **Selective:** STAR-Trap does not bind free IL-6, thereby preserving classical IL-6 signaling via the membrane-bound IL-6R. **Competitive:** STAR-Trap binds the IL-6/sIL-6R complex with high affinity, blocking its interaction with gp130 and inhibiting IL-6 trans-signaling. **Immunomodulatory:** STAR-Trap-expressing macrophages efficiently internalize the inflammatory complex through receptor-mediated endocytosis and deliver it to lysosomes for degradation.



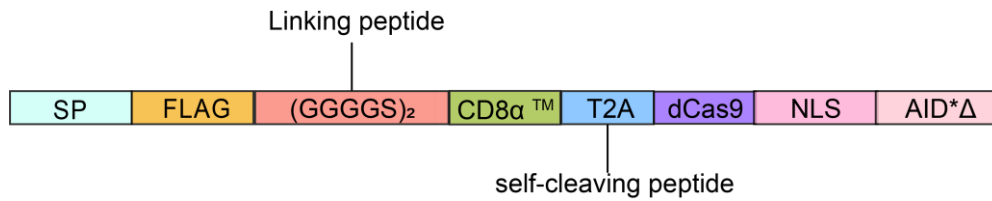
Extended Fig 2. Preparation of fluorescently labeled sIL-6R/IL-6 complex for binding and uptake assays.

The sIL-6R/IL-6 complex was first biotinylated, then incubated with streptavidin–APC. Streptavidin contains four biotin-binding sites, allowing it to simultaneously bind biotinylated complex and carry APC fluorophores, thereby generating a fluorescently labeled sIL-6R/IL-6 complex. The labeled complex was subsequently co-cultured with Raji B cells to assess binding and internalization by flow cytometry.



Extended Fig 3. Molecular dynamics simulation of candidate complexes in solution.

A 100-ns MD simulation was performed for the three candidate complexes and the gp130 reference complex. The buried surface area (nm²) is plotted against time (ns). A2B3C3 (light blue), A2B3C2 (yellow), A3B2C2 (red), and gp130 (gray) are shown. A2B3C3 maintained an extensive and stable binding interface throughout the simulation, with the lowest receptor-side root-mean-square deviation (RMSD; 1.34 ± 0.24 Å) and the largest mean buried surface area (31.82 ± 1.73 nm²).



Extended Fig 4. Schematic of the dCas9-AID*Δ expression cassette.

The cassette sequentially encodes a CD8 signal peptide (CD8 SP), a FLAG tag, a (GGGGS)₂ linking peptide, the CD8α transmembrane domain (CD8α TM), a T2A self-cleaving peptide, dCas9, a nuclear localization signal (NLS), and the hyperactive AIDΔ. In this design, the CD8 SP directs the upstream CD8-derived module to the secretory pathway for cell-surface presentation; the FLAG tag enables immunological detection; the (GGGGS)₂ linker provides flexibility between the tag and the transmembrane domain; the CD8α TM anchors the upstream module on the plasma membrane as a surface marker; the T2A peptide mediates ribosomal skipping, separating the upstream surface marker from the downstream dCas9-AIDΔ mutagenesis module and allowing the two components to be expressed as separate proteins; dCas9 provides RNA-guided targeting to the desired genomic loci; the NLS directs the dCas9-AIDΔ module into the nucleus; and the hyperactive AIDΔ catalyzes cytidine deamination, introducing localized point mutations. This design was intended to increase the local residence and mutagenic activity of AID near the target regions, thereby supporting sustained generation of localized sequence diversity within the A2, B3, and C3 coding regions.

Extended Data

Extended Data Table 1. Candidate sequences.

ID	Binder PTM	minimum inter- chain pAE	target- aligned binder RMSD	statistical potential	MM/GBSA	MM/PBSA	Binder Seq
part1_1p9m_11_model_9_1	0.83	9.47	2.11	-165.69	16.01	-30.18	MELIKLLKEIYKVVKDKEQKALILLIGEEYYKLGKL EEA EKYFKEALKLAKEAEKEGELMKMIKESLEKVEKEK MIENLKKLIEAAKKAKDPNLGYQYLVLIDFLLKENK
part1_1p9m_0_model_5_1	0.81	12.98	3.12	-97.42	8.92	-28.55	KEEA KEVLEEAKKDPNVNEEDKKKMEELLK SHPVEEKLEEVLLKKLEELEKEEEKKLKEELEKAKEKQ
part1_1p9m_12_model_3_1	0.64	3.53	7.38	-188.66	3.46	-32.58	KENP EEAKKKAEELEKEIKKLEEEGEKLIIEELKK EEELQKEVAEILPKYMKEKKDYNEALKAAAEDLNK
part1_1p9m_3_model_4_1	0.7	21.46	4.05	-122.01	1.08	-29.86	EWVKK TPYKLVYDEETNEIVVVPKS KRKEEIKKLVKELLEAVKNGDEEKAEEKLGKKLKELV
part1_1p9m_36_model_2_1	0.81	14.02	5.93	-154.41	0.66	-33.2	PKEQ YDKIEELVVKYEIAYKEGDEEAMKKYLKELEKAVEE DVETAVKVQTEILLDQLEEILKKKIEEGNPPEAKPKC
part1_1p9m_36_model_8_1	0.78	12.34	6.92	-135.24	-2.51	-44.28	EEA KKKLEEAKKKALKIENPEEAVKYLDELAKEELEKLE ELKELEKEISEEIEKLWDILLKIDVEKMAELQEKMMD
part1_1p9m_32_model_1_1	0.78	15.56	9.98	-133.44	-3.15	-33.6	LSG IEAVEYLIIEYVSSLIDTEEVKKIHKLLKELLKKIKEL DEEEIKKLLLEEAKKALEEKKYEEAEKLAREVIEKSTD
part1_1p9m_6_model_1_1	0.8	15.83	6.12	-151.24	-3.51	-55.56	KEL IEEAAEILLIARLALAKSAEEAIEAVEELEKLVKEKL SEEKEKLLIEEKLKEAEELLKENVPAEVVIKMAEYELS
part1_1p9m_9_model_2_1	0.77	2.93	2.69	-230.48	-5.37	-28.91	KNP

ID	Binder PTM	minimum inter- chain pAE	target- aligned binder RMSD	statistical potential	MM/GBSA	MM/PBSA	Binder Seq
part1_1p9m_28_model_2_1	0.75	10.74	7.76	-161.27	-5.97	-52.32	LLSEEDVKKAIERLKEKIEELE YEELAKKLKEKGF EKAAEVLEEYLKKGKLEEEAKKE AEKAA AEELEEERKKALS IKDEEEREKALEEVLKKEEELEKL L
part1_1p9m_24_model_6_1	0.71	6.87	1.64	-220.47	-6.3	-54.42	MEEAEKVAKKIVESALGPEIKKLEEKVKEYTEKGDF ENAE KAKKELEEKKKKYEEELKKLLEEVKELL KELLKKKAEEAKKKA EELKKKVEELKKKGTD E SLEE
part1_1p9m_6_model_4_1	0.71	2.71	3.91	-149.72	-6.88	-26.13	AIKL RLEAEAEKEFAKKLEELAKTVSEEEAE EAEIKLKKEL TRLLPVDEAAIAALVDEELAPIRASLATADPATAAAY LAE
part1_1p9m_24_model_5_1	0.7	11.06	6.4	-152.07	-8.88	-36.51	LAAEGRAPLEAAAAAARAAGLTEFEVEL SMVEKQYEEIHELCKELNKP ELAEKVEKLKKEALEAE KSR
part1_1p9m_30_model_0_1	0.72	13.75	7.85	-202.51	-9.32	-39.9	DPEALIAKLEEMEKLYEEVKKLKKEE KIEEYIEKLKKLEEELKKAAEEEGNEEKIKKLKKEKEE LLK
part1_1p9m_28_model_7_1	0.63	4.43	2.12	-172.35	-12.57	-53	KKDEIAGEAAVELIKKMAKEGKDPEEIVEAIEEMEKK L MEEEFKKFEEELSEELFEKYKELAVKGKEEEALELEK KEF
part1_1p9m_38_model_0_1	0.72	15.94	3.27	-143.36	-14.32	-46.61	LKKVEEYLKEKGFSEEEVEKKIKELEENL MLEKAREFAIKILEKDGIPYANEEMIEALVEEVLKILE KN
part1_1p9m_25_model_4_1	0.7	15.94	3.27	-148.92	-17.03	-34.55	PNATFEEVYEK ENEKKAKEAYEKGKKYLEEKKYDEA IKYFEEAIKLN PNSK
part1_1p9m_27_model_5_1	0.84	5.69	9.83	-169.4	-17.06	-39.71	

ID	Binder PTM	minimum inter- chain pAE	target- aligned binder RMSD	statistical potential	MM/GBSA	MM/PBSA	Binder Seq
part1_1p9m_23_model_1_1	0.72	14.29	5.07	-164.08	-17.56	-53.94	IASESYyliadiYMKEKKLPEALELLLKANAIPNQ EVNNHPEVKKLKEELEKAKAEGDEEKAKKLEEELEK KSFE LELLSLGLSKEEAekLAELYIESLkKEKE EKGEKLLKEAEEHAKKAeELKKEAKKTKDPKKKEE LLKKA KEHLEKAEEALLKAMVYYPEEYEELLKKLEELEKI EVEELVKEAKKLMEeKKYDEAIEKIKKAVEKSDDPEI KDL YYEALLRAGKVEEAQEFAEKNI KELSLFGKALLKCLETEKLPSEEIfDRAVVEAAFALS KER EGVIEALCKMTGASREEAEKLIKKLGyKF SALEELLKKEAEELAKKFQEKYPVPNLEKVKEKVYK KLKE LEKKNEEISEEELIEELKKEL EVEELLKKIEELEEKFLERVrKAKDPEEREKIKEEARE EL KPLLEKLAELLGGSPGEFTEKQLELAeERLERVWKES SKVEELVKRIEEKEKELEKELEELKKKYPELTeeELE KIK EERSEeIKELIDEATKES AGEQLEKFLEAIEKINEAIKKKDVEEVKKYMKELEKL REE VLEKLGPNSelVELAEKTLKEAKEYIKKEEEK ALEKAKELIKEALKLEEEGKYEEAIKLYKEAAELDPS LKG ECLFLISLCyLKLgKRKEAAESLEKALPDLVKESPYDI
part1_1p9m_27_model_2_1	0.75	6.95	5.26	-220.85	-17.65	-51.5	
part1_1p9m_17_model_9_1	0.81	8.32	6.57	-193.13	-18.23	-33.6	
part1_1p9m_38_model_5_1	0.73	9.18	1.58	-241.22	-18.87	-42	
part1_1p9m_29_model_5_1	0.77	5.52	6.98	-165.91	-19.82	-64.44	
part1_1p9m_6_model_5_1	0.7	10.25	6.9	-240.45	-21.11	-76.57	
part1_1p9m_14_model_4_1	0.73	1.35	3.02	-282.39	-21.22	-62	
part1_1p9m_31_model_1_1	0.75	18.22	9.19	-131.93	-21.87	-64.73	
part1_1p9m_28_model_3_1	0.78	10.48	5.16	-245.35	-22.51	-53.79	

ID	Binder PTM	minimum inter- chain pAE	target- aligned binder RMSD	statistical potential	MM/GBSA	MM/PBSA	Binder Seq
part1_1p9m_21_model_2_1	0.71	15.74	6.87	-197.21	-23.6	-41.08	MEELLKKAKEAIEKGNKEEVEKIVEEVIEKGDLESLL EIY KECEKYGIE
part1_1p9m_31_model_4_1	0.76	13.36	2.27	-240.2	-27.15	-54.74	AAAEVEAAVARYEAAAERYLAACKRAGAITPEQEREL ARAY NELVALGDPAAAAARLAELDARFAAEEAAAAAAA ELEKIFKEYKEIHKYKEGKLTKEEYEKELDKLLEEIQ KL
part1_1p9m_29_model_7_1	0.78	16.37	3.84	-200.94	-31.61	-50.21	LKDDIEKQIEAFEELEKIEKE MTVEELKEEIEKLEKEIEAAKAAGKPAAEAAHLQEKL EVLK
part1_1p9m_39_model_6_1	0.79	1.88	5.06	-261.84	-32.26	-73.88	SFLKELEEDKENKEALSISLRGTLALIDKEAAGEL MDVEKKIEEIEKIEKEAEKAKKEAEKYKEEAKKLKE EKAP
part1_1p9m_13_model_2_1	0.71	12.53	6.36	-152.09	-34.32	-57.06	EEKVKELEKKAKEAEFEAKKLEEEIFKYLKEAAV MELEKLIKEIFEKVEKYLKLKKENASEEELKKLEEEIE KL
part1_1p9m_24_model_7_1	0.81	1.47	5.53	-237.87	-34.34	-75.9	VEEAYKAAKKEGKLAKFQEALLEIDKLL TKEEELKKEALKAASFIVENKKIAPDLVDEKLEKALK DYY
part1_1p9m_6_model_8_1	0.77	1.32	1.55	-286.37	-34.9	-62.42	IKEISEKFGKPKEEVEEIEKEIEKLEKIGGKVTEEA MEEELKKKAEEELAKKAEEAREKGDKKKAIEYLEELL KVLN
part1_1p9m_20_model_8_1	0.82	4.7	4.08	-262.87	-37.38	-77.71	EIKTDPEKEEKIKIAEVELAWEKGDWKAYLEALEKL MIEEEAEKLAKEELEKVYEEKFKELVLKGPEFAIEEV EKL
part1_1p9m_20_model_5_1	0.75	4.19	3.67	-232.63	-38	-56.23	AKEEAKKKIIEYLIIEKKIPKEEAEDIAEKVLEKLK KEELEKEMEKIMKEIEEAEEKGDKEKARELMKKLIEL EEK
part1_1p9m_37_model_7_1	0.72	1.2	7.38	-252.22	-40.7	-72.52	

ID	Binder PTM	minimum inter- chain pAE	target- aligned binder RMSD	statistical potential	MM/GBSA	MM/PBSA	Binder Seq
part1_1p9m_0_model_6_1	0.65	1.86	1.45	-279.13	-43.78	-76.75	YLK IDPEKLKEEIKKLKEEKLEEFEPKYKELIKELAKKNLP EE EVLKEVEKFYKEFEKEFEKEIEKLKK SKEIEKEAEKAFEEKKYEEAIKLYEEALKKDPSKKLY
part1_1p9m_4_model_3_1	0.83	3.1	2.72	-255.47	-50.57	-70.4	YKE GIAFAYYELGKKEEARKLFIEIAEEL DLEVEMLKRLAEEAEKEGNWEELEGIGLELKKLGYE
part1_1p9m_0_model_7_1	0.73	14.78	4.69	-195.82	-50.67	-78.66	EEAK KVLERAIKGYKEKGLEEKAKKVEEIL ILEKIKEYIKEKLRKGEVISLEELWKIAKEKFGITKEEA
part1_1p9m_20_model_1_1	0.79	3.08	5.91	-207.5	-53.73	-90.38	E EELKEAHEEAVFEVIREKAKEGASVEELIEFLEKAE EEEEKRKLKREYQELIIEYEKAEEEGKIEEAVKLLEEI
part1_1p9m_4_model_0_1	0.77	17.47	1.43	-207.73	-56.17	-92.92	LK LAEKLGLPREEIKSIKEELEKLKKKI LAELYGAAQRALAEERYEEAIALLEEILARREEVPPS
part2_trp249_1p9m_33_model_0_1	0.77	17.68	9.14	-78.43	-0.23	-11.62	LL AAAERLLAEAREKL ETVTTEVVVEEPDRLTITRTVSGVAEPPPERIEELQRAL
part2_trp249_1p9m_19_model_3_1	0.75	18.74	3	-155.85	-3.89	-25.71	EA EVRARGFEPVLVRAVVENGVLTVTAVGVRPL MKKAEVEIYDEDGTLVHSEVVPVAEDGTVTYTFTVP
part2_trp249_1p9m_25_model_4_1	0.79	19.03	5.06	-105.19	-8.68	-27.21	ES GKYTVKVYDEFGRLVYEETIEAKAGETYTVTVKVEP EREALVERAVARYEEVRALLEAGDLEGFEEAIRRIAE
part2_trp249_1p9m_19_model_7_1	0.84	20.78	6.68	-95.55	-9.54	-24.2	EL GEAVATTIGALARNRAPAEEARRVIA RVVAAL

ID	Binder PTM	minimum inter- chain pAE	target- aligned binder RMSD	statistical potential	MM/GBSA	MM/PBSA	Binder Seq
part2_trp249_1p9m_34_model_4_1	0.71	12.18	5.76	-109.53	-9.99	-32.44	PELVGEAVLDGRRLRLYLRETKDEQGKPVWELEEFADG ELVRRRTFASREEALAHVARLRG
part2_trp249_1p9m_28_model_7_1	0.72	16.33	7.38	-96.18	-12.79	-35.54	LYEAVVTWEKNGQKGRAVRVYPSREEAEAAAEIEIR RRL EAEGATNIRVEIR
part2_trp249_1p9m_8_model_4_1	0.77	15.29	2.54	-164.05	-15.36	-24.36	VVRSASALAKTPEEALAKAKAKAEALGFKVLEVAS LV KVGDKWQATVRGEKEV
part2_trp249_1p9m_14_model_7_1	0.71	21.44	3.48	-86.01	-15.87	-33.74	VLEVAAYSVSADGKTVVVNINQTAAALGISVEEAVARL TAAVPSTVIETAPDGTVYGRVPAAAFAA
part2_trp249_1p9m_36_model_5_1	0.74	7.73	9.36	-177.24	-16	-30.91	KTVERVVGEVVFAGTPAEERLAAVEALAAERAPAK PA RLRMELVEVLPDGSVRYRVIAIYEL
part2_trp249_1p9m_32_model_0_1	0.85	21.6	7.88	-131.7	-16.13	-25.69	QREAVVEELERYGISRELAERLVEEAAALGAASAVVSPE EAVDIASIHQTLAATLPGEEAVREALRRAAEIIAEIE AE
part2_trp249_1p9m_9_model_0_1	0.77	17.03	4.04	-149.91	-20.7	-33.96	PEVRTETITSDLPPEEVRKEVAARIAELKAQGAKEVV VID QRVEGSSTVTTLLEVTY
part2_trp249_1p9m_6_model_6_1	0.75	4.4	2.1	-119.54	-21	-34.72	RVVVEREVIEFEEPIIEEARPIAERRVEELERRLREEY PD AEIEVRASFIEEDGLVKELIVEAVVRVP
part2_trp249_1p9m_30_model_1_1	0.71	18.45	3.24	-133.3	-22.21	-39.46	ATAVVTVDSPRPRTVDIECVDPGIRLRATYPPGRHERTF EVPPGSVIRFLDPANAEGVTVTVTTE

ID	Binder PTM	minimum inter- chain pAE	target- aligned binder RMSD	statistical potential	MM/GBSA	MM/PBSA	Binder Seq
part2_trp249_1p9m_39_model_0_1	0.72	16.42	1.32	-137.14	-24	-32.83	PRLRTLTYYPKEKAEERIARIVAALASGLRVVDVK REQ VDEDTVRLTITVEE VTVRVGPFATEEEAVAAAEAGREVAARLGGRLAEI GLV RGEAWTGVAVVEVP RELEQVIVLPPDAELLSVSGDCTVEVEELPDGTTRVT VRY REGQTFTITYRYRDDVVTETFSV DTLARQEAVEAFLRARGLEPQQIELRLEPDGSWVAV AYV DAGPGRLLRVTVREVPGGGWEVVAEELLELP PCVLPPPFTLTVLFPGDAEPRTWLCTPVRCFTEDGVP LAE YACRSLEDPSLTGKAVLRLEGDEWVVSVSYNRPEP PGRTLTITLVITVSGAESVEAARAAALAQLAALKPGT TIT IAGRPYRLLSVGEYTVIEDPKTKSATVEVEAVVQEL AAELRELAALVEAAFPEWRVVGFEEDGELCVVAL REL PDGRLLRALFRRREDGSVELVRVEEVPL APRRYVITDDRELMEEILEADPSVVAITVESREAAER LAA RLRAVPEDVADVEVRVRP MLKKYLVVARPVEGNQPSIEIEVEAASPEEALEKAIEI LRE QGITGTWVISVIVDGELVERRTVTIE
part2_trp249_1p9m_33_model_9_1	0.77	14.91	7.33	-124.44	-24.7	-38.2	
part2_trp249_1p9m_36_model_1_1	0.73	16.63	7.61	-175.33	-26.15	-34.45	
part2_trp249_1p9m_29_model_7_1	0.78	7.72	1.64	-183.09	-27.84	-44.56	
part2_trp249_1p9m_31_model_0_1	0.73	23.23	4.13	-172.53	-28.95	-40.2	
part2_trp249_1p9m_31_model_8_1	0.74	20.85	5.04	-196.08	-29.22	-46.4	
part2_trp249_1p9m_16_model_5_1	0.72	23.32	8.89	-89.3	-29.49	-41.64	
part2_trp249_1p9m_7_model_3_1	0.72	19.1	3.57	-177.08	-30.3	-33.29	
part2_trp249_1p9m_23_model_7_1	0.79	17.61	4.26	-125.66	-31.22	-42.88	

ID	Binder PTM	minimum inter- chain pAE	target- aligned binder RMSD	statistical potential	MM/GBSA	MM/PBSA	Binder Seq
part2_trp249_1p9m_12_model_4_1	0.78	5.27	1.52	-225.85	-36.16	-55.74	MTPEQLAKADEYFAKSEEARKKAGEYLKLYDETGD PALLEKAEAYRKEADIYTAKGYKAAGDTETAKAIAD KLGVTL
part2_trp249_1p9m_22_model_4_1	0.81	19.44	8.87	-162.69	-38.14	-43.07	GMKRMRETITVAPEKVPEAVALLEELKARLPEARIEF AVDGNITITVEVEFPPEEALPRLEEALAEFYARIRPLSRR REYES
part2_trp249_1p9m_38_model_8_1	0.78	5.66	1.32	-231.18	-39.91	-55.54	REEEFEALVAEHWP EIEAVYAEAARTLSREETAAAV RAAIEALLPPEVAAYAIAHYAPKLE
part2_trp249_1p9m_2_model_7_1	0.78	3.74	2.46	-237.48	-47.69	-61.14	MLKFLGLSITPAPQTEEEYEKVEKLQKEVEEKHKELS KKLDKETTLKELKKYAEEKAKELKLPAKRINIHVEE GE
part2_trp249_1p9m_10_model_5_1	0.78	9.15	1.77	-164.51	-49.62	-65.61	MEVVERTLEIDAAAP EEEAVARVLAALPALPEALAR AAVREVKANPGSEVTVG VETPERRELLTVVSR
part2_trp249_1p9m_19_model_9_1	0.82	8.27	1	-238.06	-56.44	-62.69	RLDEL VASYAAHLAELGLAPEEVAAA VARIRAAAAA ARAANASLPDPRLRERAVLDAASAEIERIGRERAA
part2_trp249_1p9m_6_model_2_1	0.74	15.29	1.78	-192.2	-60.73	-72.8	SSKVLRYRRVGDADIDELIATVKEQLAAEGASDIRLL ERSPDGKR VVLVYKDPNGEERYAIVEVVEEEP
part2_trp249_1p9m_35_model_5_1	0.76	18.32	1.63	-326.46	-62.66	-74.16	SAALRARGLAALRAEAERLGLDPAAVRLVAVLRVG DRVVALGELPPEAYPHADPDDSRVRFEAEVPA
part2_trp249_1p9m_31_model_3_1	0.78	20.29	2.86	-127.65	-64.53	-59.88	IPANVQVGLPDPSLLPLALKAVEVARKVVGP DATIGV TYPNETKTF SVHAQAPGQP PRYLIVAEDGTVLEEGEV PP
part2_trp249_1p9m_5_model_3_1	0.64	1.67	1.76	-274.29	-67.9	-69.43	SVKVVT SQCTLPDGT VVTVTVSCTGPGEYTWEVSFY DADDPSNRTPPCTVTCTCSPDDP
part2_trp249_1p9m_6_model_7_1	0.8	10.14	1.7	-231.37	-70.53	-83.02	APSAEQLAALAAEARAALLAQAAELGWTPSDLELEV RPVPEGVLA IARATATRP GQPPVRLRAEALLRL
part2_trp296_1p9m_38_model_5_1	0.72	9.53	3.32	-128.66	-24.38	-30.51	ARHTLPLTVPEGETDVYVDGKLVATVPAGEQEVTVE VPPGTHRVS LRNRKTGESRSSDVVFP

ID	Binder PTM	minimum inter-chain pAE	target-aligned binder RMSD	statistical potential	MM/GBSA	MM/PBSA	Binder Seq
part2_trp296_1p9m_25_mo del_8_1	0.71	22.82	8.66	-164.27	-32.59	-44.19	GAAARQAAALDAAAEAAEAAGDLAAAAAYAAAA ALLAAHAATHPNGANLAALAAEAAARAAALAALA A
part2_trp296_1p9m_29_mo del_0_1	0.72	22	3.84	-198.33	-65.43	-74.04	GFRAAVPLRVGDPVEIRLDDGTVPGRVDRVEGDVV SVAPLEPLTPGVHRVTVRFLTEPPQTSEPRLVEFPFRV P

After sorting by the average score per residue and selecting the top 1 sequence, a total of 78 candidate sequences were obtained. ID, binder identifier. Binder PTM, predicted post-translational modification score. Minimum inter-chain PAE, minimum inter-chain predicted aligned error. Target-aligned binder RMSD, RMSD of the binder aligned to the target (Å). Statistical potential, statistical potential score of the complex. MM/GBSA and MM/PBSA, binding free energies calculated by MM/GBSA and MM/PBSA methods (kcal/mol), respectively. Binder Seq, amino acid sequence of the designed binder.

Extended Data Table 2. 27 candidate sequences constructed by homologous recombination of three segments.

ID	Sequence
A1B1C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TGGAGGAAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCCAGGGAAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCT CGAGGAGCTGCTGAAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAAGATTGCTGAGGTGGAGTTGGCATGGGAAAAGG GCGATTGGAAGGCCTATCTGGAGGCTCTCGAGAAACTCGACTACAAGGACGACGACGACAAGTCCGCCGCATTGCGCGCGAGAGGACTGGCCGCTC TCAGAGCAGAAGCCGAAAGGCTGGGCCTCGACCCTGCAGCAGTCAGGCTCGTGGCAGTGCTGCGGGTGGGAGACAGGGTGGTGGCTTTGGGAGAG CTGCCCCCAGAAGCATACCCCCACGCTGACCCAGACGATAGCAGAGTGCCTTCGAGGCTGAGGTGCCCCGCTGGCGGGGGCAGCGGCGGGGGCTCC GCCCCGCATACTCTGCCCCTGACCGTGCCTGAAGGCGAAACAGACGTATACGTGGATGGAAAGCTGGTGGCCACAGTGCCTGCTGGGGAGCAGGAG GTGACGGTGGAAAGTGCCCCCGGAACCCATCGCGTGAGCCTCCGCAATCGGAAGACTGGCGAGAGCAGAAGCTCCGATGTGGTCTTCCCTATCATTG TGGCGGTGGTGAT

ID	Sequence
A1B1C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TGGAGGAAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCCAGGGAAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCT CGAGGAGCTGCTGAAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAAGATTGCTGAGGTGGAGTTGGCATGGGAAAAGG GCGATTGGAAGGCCTATCTGGAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGTCCGCCGCATTGCGCGCGAGAGGACTGGCCGCTC TCAGAGCAGAAGCCGAAAGGCTGGGCCTCGACCCTGCAGCAGTCAGGCTCGTGGCAGTGCTGCGGGTGGGAGACAGGGTGGTGGCTTTGGGAGAG CTGCCCCCAGAAGCATACCCCCACGCTGACCCAGACGATAGCAGAGTGCGCTTCGAGGCTGAGGTGCCCCTGCGGGGGGACGCGGCGGGGGCTCC GGAGCCGCCGCCAGACAAGCCGCAGCCCTGGACGCCGCCGCCGAAGCAGCCGAGGCTGCGGGAGACTTGGCCGCAGCTGCAGCAGCATATGCAGC CGCTGCAGCTCTCCTGGCAGCTCACGCCGCCACTCATCCGAATGGCGCCAACTTGGCCGCTTTGGCTGCTGAAGCCGCCGCTAGAGCCGCCGCCCTG GCTGCTCTGGCGGCCATCATTGTGGCGGTGGTGAT
A1B1C3	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TGGAGGAAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCCAGGGAAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCT CGAGGAGCTGCTGAAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAAGATTGCTGAGGTGGAGTTGGCATGGGAAAAGG GCGATTGGAAGGCCTATCTGGAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGTCCGCCGCATTGCGCGCGAGAGGACTGGCCGCTC TCAGAGCAGAAGCCGAAAGGCTGGGCCTCGACCCTGCAGCAGTCAGGCTCGTGGCAGTGCTGCGGGTGGGAGACAGGGTGGTGGCTTTGGGAGAG CTGCCCCCAGAAGCATACCCCCACGCTGACCCAGACGATAGCAGAGTGCGCTTCGAGGCTGAGGTGCCCCTGCGGGGGGACGCGGCGGGGGCTCC GGCTTTAGAGCAGCTGTCCCACTGAGAGTCGGGGACCCGGTCGAAATTAGGCTGGACGACGGGACAGTGGTTCCAGGCAGGGTCGATCGCGTGGAG GGCGACGTGGTCTCCGTGGCCCCCTCTTGAGCCACTACCCCCGGGCGTGACCCGGGTGACCGTGAGGTTCCCTGACCGAGCCTCCGCAGACCAGCGAG CCCAGACTGGTGGAGTTTCCCCCCCCGGGTTCCCATCATTGTGGCGGTGGTGAT
A1B2C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TGGAGGAAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCCAGGGAAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCT CGAGGAGCTGCTGAAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAAGATTGCTGAGGTGGAGTTGGCATGGGAAAAGG GCGATTGGAAGGCCTATCTGGAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGCACCCTCCC AGATGGCACAGTGGTGACCGTAACTGTCTCTTGACCCGGACCCGGCGAGTACACGTGGGAGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGC ACCCCTCCTTGACCCGTGACTTGACCTGTTCCCCCGACGACCCGGCGGGGGGACGCGGCGGGGGCTCCGCCCGCCATACTCTGCCCCTGACCGTGC CTGAAGGCGAAACAGACGTATACGTGGATGGAAAGCTGGTGGCCACAGTGCCTGCTGGGGAGCAGGAGGTGACGGTGAAGTGCCCCCGGAACC CATCGCGTGAGCCTCCGCAATCGGAAGACTGGCGAGAGCAGAAGCTCCGATGTGGTCTTCCCTATCATTGTGGCGGTGGTGAT

ID	Sequence
A1B2C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TGGAGGAAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCCAGGGAAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCT CGAGGAGCTGCTGAAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAAGATTGCTGAGGTGGAGTTGGCATGGGAAAAGG GCGATTGGAAGGCCTATCTGGAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGCACCCTCCC AGATGGCACAGTGGTGACCGTAACTGTCTCTTGCACCGGACCCGGCGAGTACACGTGGGAGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGC ACCCCTCCTTGCACCGTGACTTGCACCTGTTCCCCCGACGACCCGGCGGGGGCAGCGGCGGGGGCTCCGGAGCCGCCGACAGACAAGCCGCAGCC CTGGACGCCGCCGCCGAAGCAGCCGAGGCTGCGGGAGACTTGGCCGCAGCTGCAGCAGCATATGCAGCCGCTGCAGCTCTCCTGGCAGCTCACGCC GCCACTCATCCGAATGGCGCCAACTTGGCCGCTTTGGCTGCTGAAGCCGCCGCTAGAGCCGCCGCCCTGGCTGCTCTGGCGGCCATCATTGTGGCGG TGGTGAT
A1B2C3	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TGGAGGAAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCCAGGGAAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCT CGAGGAGCTGCTGAAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAAGATTGCTGAGGTGGAGTTGGCATGGGAAAAGG GCGATTGGAAGGCCTATCTGGAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGCACCCTCCC AGATGGCACAGTGGTGACCGTAACTGTCTCTTGCACCGGACCCGGCGAGTACACGTGGGAGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGC ACCCCTCCTTGCACCGTGACTTGCACCTGTTCCCCCGACGACCCGGCGGGGGCAGCGGCGGGGGCTCCGGCTTTAGAGCAGCTGTCCCACTGAGA GTCGGGGACCCGGTCGAAATTAGGCTGGACGACGGGACAGTGGTTCCAGGCAGGGTCGATCGCGTGGAGGGCGACGTGGTCTCCGTGGCCCCCTCTT GAGCCACTCACCCCGGGCGTGACCGGGTGACCGTGAGGTTCTTGACCGAGCCTCCGCAGACCAGCGAGCCCAGACTGGTGGAGTTTCCCCCGCG GTTCCCATCATTGTGGCGGTGGTGAT
A1B3C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TGGAGGAAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCCAGGGAAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCT CGAGGAGCTGCTGAAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAAGATTGCTGAGGTGGAGTTGGCATGGGAAAAGG GCGATTGGAAGGCCTATCTGGAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCG CAGAAGCCAGAGCTGCTTTGCTGGCCCAGGCAGCCGAACTGGGCTGGACTCCGAGCGATCTCGAGTTGGAGGTGCGGCCAGTCCCCGAGGGAGTGC TGGCAATCGCACGCGCTACAGCCACCAGACCTGGCCAACCGCCCGTGAGATTGAGAGCCGAGGCCCTCCTGCGCCTGGGCGGGGGCAGCGGCGGG GGCTCCGCCCCGCCATACTCTGCCCCTGACCGTGCCTGAAGGCGAAACAGACGTATACGTGGATGGAAAGCTGGTGGCCACAGTGCCTGCTGGGGAG CAGGAGGTGACGGTGAAGTGCCCCCGGAACCATCGCGTGAGCCTCCGCAATCGGAAGACTGGCGAGAGCAGAAGCTCCGATGTGGTCTTCCCT ATCATTGTGGCGGTGGTGAT

ID	Sequence
A1B3C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TGGAGGAAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCCAGGGAAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCT CGAGGAGCTGCTGAAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAAGATTGCTGAGGTGGAGTTGGCATGGGAAAAGG GCGATTGGAAGGCCTATCTGGAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCG CAGAAGCCAGAGCTGCTTTGCTGGCCCAGGCAGCCGAACCTGGGCTGGACTCCGAGCGATCTCGAGTTGGAGGTGCGGCCAGTCCCCGAGGGAGTGC TGGCAATCGCACGCGCTACAGCCACCAGACCTGGCCAACCGCCCGTGAGATTGAGAGCCGAGGCCCTCCTGCGCCTGGGCGGGGGCAGCGGCGGG GGCTCCGGAGCCGCCGCCAGACAAGCCGCAGCCCTGGACGCCGCCGCCGAAGCAGCCGAGGCTGCGGGAGACTTGGCCGCAGCTGCAGCAGCATA TGCAGCCGCTGCAGCTCTCCTGGCAGCTCACGCCGCCACTCATCCGAATGGCGCCAACCTTGGCCGCTTTGGCTGCTGAAGCCGCCGCTAGAGCCGCC GCCCTGGCTGCTCTGGCGGCCATCATTGTGGCGGTGGTGAT
A1B3C3	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TGGAGGAAGAACTGAAGAAAAAGGCTGAGGAGTTGGCCAAAAAGGCTGAGGAGGCCAGGGAAAAAGGGTGACAAGAAGAAGGCCATCGAGTACCT CGAGGAGCTGCTGAAGGTGCTGAACGAAATCAAGACAGATCCAGAGAAGGAGGAGAAGATCAAGATTGCTGAGGTGGAGTTGGCATGGGAAAAGG GCGATTGGAAGGCCTATCTGGAGGCTCTCGAGAACTCGACTACAAGGACGACGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCG CAGAAGCCAGAGCTGCTTTGCTGGCCCAGGCAGCCGAACCTGGGCTGGACTCCGAGCGATCTCGAGTTGGAGGTGCGGCCAGTCCCCGAGGGAGTGC TGGCAATCGCACGCGCTACAGCCACCAGACCTGGCCAACCGCCCGTGAGATTGAGAGCCGAGGCCCTCCTGCGCCTGGGCGGGGGCAGCGGCGGG GGCTCCGGCTTTAGAGCAGCTGTCCCACTGAGAGTCGGGGACCCGGTCGAAATTAGGCTGGACGACGGGACAGTGGTTCCAGGCAGGGTCGATCGC GTGGAGGGCGACGTGGTCTCCGTGGCCCCCTTTGAGCCACTCACCCCGGGCGTGACCCGGGTGACCGTGAGGTTCTTGACCGAGCCTCCGCAGACC AGCGAGCCCAGACTGGTGGAGTTTCCCCCCCCGGGTTCCTCATCATTGTGGCGGTGGTGAT
A2B1C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TTGACCCTGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAGAATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTG GCTAAGAAGAACCCTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTTCGAGAAAGAGTTTCGAGAAGGAGATTGAGAAGCT CAAAAAAGACTACAAGGACGACGACGACAAGTCCGCCGCATTGCGCGCGAGAGGACTGGCCGCTCTCAGAGCAGAAGCCGAAAGGCTGGGCCTCG ACCCTGCAGCAGTCAGGCTCGTGGCAGTGCTGCGGGTGGGAGACAGGGTGGTGGCTTTGGGAGAGCTGCCCCCAGAAGCATAACCCACGCTGACC CAGACGATAGCAGAGTGCGCTTCGAGGCTGAGGTGCCCCTGGCGGGGGCAGCGGCGGGGGCTCCGCCCGCCATACTCTGCCCTGACCGTGCCTG AAGGCGAAACAGACGTATACGTGGATGGAAAGCTGGTGGCCACAGTGCCTGCTGGGGAGCAGGAGGTGACGGTGGAAAGTGCCCCCGGAACCCAT CGCGTGAGCCTCCGCAATCGGAAGACTGGCGAGAGCAGAAGCTCCGATGTGGTCTTCCCTATCATTGTGGCGGTGGTGAT

ID	Sequence
A2B1C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TTGACCCTGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAGAATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTG GCTAAGAAGAACCTTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTCGAGAAAGAGTTCGAGAAGGAGATTGAGAAGCT CAAAAAAGACTACAAGGACGACGACGACAAGTCCGCCGCATTGCGCGCGAGAGGACTGGCCGCTCTCAGAGCAGAAGCCGAAAGGCTGGGCCTCG ACCCTGCAGCAGTCAGGCTCGTGGCAGTGCTGCGGGTGGGAGACAGGGTGGTGGCTTTGGGAGAGCTGCCCCCAGAAGCATAACCCCCACGCTGACC CAGACGATAGCAGAGTGCGCTTCGAGGCTGAGGTGCCCGCTGGCGGGGGCAGCGGCGGGGGCTCCGGAGCCGCCGCCAGACAAGCCGCAGCCCTG GACGCCGCCGCCGAAGCAGCCGAGGCTGCGGGAGACTTGGCCGCAGCTGCAGCAGCATATGCAGCCGCTGCAGCTCTCCTGGCAGCTCACGCCGCC ACTCATCCGAATGGCGCCAACCTTGGCCGCTTTGGCTGCTGAAGCCGCCGCTAGAGCCGCCGCCCTGGCTGCTCTGGCGGCCATCATTGTGGCGGTGG TGAT
A2B1C3	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TTGACCCTGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAGAATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTG GCTAAGAAGAACCTTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTCGAGAAAGAGTTCGAGAAGGAGATTGAGAAGCT CAAAAAAGACTACAAGGACGACGACGACAAGTCCGCCGCATTGCGCGCGAGAGGACTGGCCGCTCTCAGAGCAGAAGCCGAAAGGCTGGGCCTCG ACCCTGCAGCAGTCAGGCTCGTGGCAGTGCTGCGGGTGGGAGACAGGGTGGTGGCTTTGGGAGAGCTGCCCCCAGAAGCATAACCCCCACGCTGACC CAGACGATAGCAGAGTGCGCTTCGAGGCTGAGGTGCCCGCTGGCGGGGGCAGCGGCGGGGGCTCCGGCTTTAGAGCAGCTGTCCCACTGAGAGTCG GGGACCCGGTCGAAATTAGGCTGGACGACGGGACAGTGGTTCCAGGCAGGGTCGATCGCGTGAGGGGCGACGTGGTCTCCGTGGCCCCCTCTTGAGC CACTACCCCCGGGCGTGACCCGGGTGACCGTGAGGTTCCCTGACCGAGCCTCCGCAGACCAGCGAGCCCAGACTGGTGGAGTTTCCCCCCCCGGGTTC CCATCATTGTGGCGGTGGTGAT
A2B2C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TTGACCCTGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAGAATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTG GCTAAGAAGAACCTTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTCGAGAAAGAGTTCGAGAAGGAGATTGAGAAGCT CAAAAAAGACTACAAGGACGACGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGCACCCTCCCAGATGGCACAGTGGTGACCGTAACTGTCTC TTGCACCGGACCCGGCGAGTACACGTGGGAGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGCACCCCTCCTTGCACCGTGACTTGACCTGT TCCCCCGACGACCCCGGCGGGGGCAGCGGCGGGGGCTCCGCCCGCCATACTCTGCCCCTGACCGTGCCTGAAGGCGAAACAGACGTATACGTGGAT GGAAAGCTGGTGGCCACAGTGCCTGCTGGGGAGCAGGAGGTGACGGTGGAAGTGCCCCCGGAACCCATCGCGTGAGCCTCCGCAATCGGAAGAC TGGCGAGAGCAGAAGCTCCGATGTGGTCTTCCTATCATTGTGGCGGTGGTGAT

ID	Sequence
A2B2C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TTGACCCTGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAGAATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTG GCTAAGAAGAACCTTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTCGAGAAAGAGTTCGAGAAGGAGATTGAGAAGCT CAAAAAAGACTACAAGGACGACGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGCACCCTCCCAGATGGCACAGTGGTGACCGTAACTGTCTC TTGCACCGGACCCGGCGAGTACACGTGGGAGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGCACCCCTCCTTGCACCGTGACTTGACCTGT TCCCCCGACGACCCCGGCGGGGGCAGCGGCGGGGGCTCCGGAGCCGCCGCCAGACAAGCCGCAGCCCTGGACGCCGCCGCCGAAGCAGCCGAGGC TGCGGGAGACTTGGCCGCAGCTGCAGCAGCATATGCAGCCGCTGCAGCTCTCCTGGCAGCTCACGCCGCCACTCATCCGAATGGCGCCAACCTTGGCC GCTTTGGCTGCTGAAGCCGCCGCTAGAGCCGCCGCCCTGGCTGCTCTGGCGGCCATCATTGTGGCGGTGGTGAT
A2B2C3	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TTGACCCTGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAGAATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTG GCTAAGAAGAACCTTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTCGAGAAAGAGTTCGAGAAGGAGATTGAGAAGCT CAAAAAAGACTACAAGGACGACGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGCACCCTCCCAGATGGCACAGTGGTGACCGTAACTGTCTC TTGCACCGGACCCGGCGAGTACACGTGGGAGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGCACCCCTCCTTGCACCGTGACTTGACCTGT TCCCCCGACGACCCCGGCGGGGGCAGCGGCGGGGGCTCCGGCTTTAGAGCAGCTGTCCCAGTGCAGAGTCGGGGACCCGGTTCGAAATTAGGCTGGAC GACGGGACAGTGGTTCCAGGCAGGGTCGATCGCGTGAGGGGCGACGTGGTCTCCGTGGCCCCCTCTTGAGCCACTCACCCCGGGCGTGCACCGGGTG ACCGTGAGGTTCTTGACCGAGCCTCCGCAGACCAGCGAGCCCAGACTGGTGGAGTTTCCCCCCCCGGGTTC CATCATTGTGGCGGTGGTGAT
A2B3C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA TTGACCCTGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAGAATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTG GCTAAGAAGAACCTTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTCGAGAAAGAGTTCGAGAAGGAGATTGAGAAGCT CAAAAAAGACTACAAGGACGACGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCGCAGAAGCCAGAGCTGCTTTGCTGGCCCAGG CAGCCGAACCTGGGCTGGACTCCGAGCGATCTCGAGTTGGAGGTGCGGCCAGTCCCCGAGGGAGTGCTGGCAATCGCACGCGCTACAGCCACCAGAC CTGGCCAACCGCCCGTGAGATTGAGAGCCGAGGCCCTCCTGCGCCTGGGCGGGGGCAGCGGCGGGGGCTCCGCCCCGCATACTCTGCCCTGACCG TGCCTGAAGGCGAAACAGACGTATACGTGGATGGAAAGCTGGTGGCCACAGTGCCTGCTGGGGAGCAGGAGGTGACGGTGGAAGTGCCCCCGGA ACCCATCGCGTGAGCCTCCGCAATCGGAAGACTGGCGAGAGCAGAAGCTCCGATGTGGTCTTCCCTATCATTGTGGCGGTGGTGAT

ID	Sequence
A2B3C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATTGACCCTGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAGAATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTGGCTAAGAAGAACCTTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTCGAGAAAGAGTTCGAGAAGGAGATTGAGAAGCTCAAAAAAGACTACAAGGACGACGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCGCAGAAGCCAGAGCTGCTTTGCTGGCCCAGGCAGCCGAACCTGGGCTGGACTCCGAGCGATCTCGAGTTGGAGGTGCGGCCAGTCCCCGAGGGAGTGCTGGCAATCGCACGCGCTACAGCCACCAGACCTGGCCAACCGCCCGTGAGATTGAGAGCCGAGGCCCTCCTGCGCCTGGGCGGGGGCAGCGGCGGGGGCTCCGGAGCCGCCGCCAGACAAGCCGCA GCCCTGGACGCCGCCGCCGAAGCAGCCGAGGCTGCGGGAGACTTTGGCCGCAGCTGCAGCAGCATATGCAGCCGCTGCAGCTCTCCTGGCAGCTCACGCCGCCACTCATCCGAATGGCGCCAACCTTGGCCGCTTTGGCTGCTGAAGCCGCCGCTAGAGCCGCCGCCCTGGCTGCTCTGGCGGCCATCATTGTGGCGGTGGTGAT
A2B3C3	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAATTGACCCTGAAAAGTTGAAGGAGGAAATTAAGAAGCTCAAGGAGGAGAAGTTGGAAGAATTCGAACCAAAGTACAAGGAGTTGATCAAGGAGCTGGCTAAGAAGAACCTTCCCGAGGAGGAGGTGCTGAAGGAGGTGGAGAAGTTCTATAAGGAGTTCGAGAAAGAGTTCGAGAAGGAGATTGAGAAGCTCAAAAAAGACTACAAGGACGACGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCGCAGAAGCCAGAGCTGCTTTGCTGGCCCAGGCAGCCGAACCTGGGCTGGACTCCGAGCGATCTCGAGTTGGAGGTGCGGCCAGTCCCCGAGGGAGTGCTGGCAATCGCACGCGCTACAGCCACCAGACCTGGCCAACCGCCCGTGAGATTGAGAGCCGAGGCCCTCCTGCGCCTGGGCGGGGGCAGCGGCGGGGGCTCCGGCTTTAGAGCAGCTGTCCCCTGAGAGTCGGGGACCCGGTCGAAATTAGGCTGGACGACGGGACAGTGGTTCCAGGCAGGGTCGATCGCGTGAGAGGGCGACGTGGTCTCCGTGGCCCCCTCTTGAGCCACTCACCCCGGGCGTGACCGGGTGACCGTGAGGTTCTTGACCGAGCCTCCGCAGACCAGCGAGCCCAGACTGGTGGAGTTTCCCCCCC GGGTTCCTCATCATTGTGGCGGTGGTGAT
A3B1C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAAGCAAAGAAATCGAGAAGGAAGCCGAGAAAGCCTTCGAGGAGAAAAAATACGAAGAGGCTATCAAAGTGTACGAGGAGGCATTGAAGAAGGACCCATCTAAGAAGCTGTACTACAAGGAAGGGATTGCCTTCGCATATTACGAAGTGGGCAAAAAAGAGGAGGCAAGGAAACTGTTTCATCGAGATCGCGGAAGAACTGGACTACAAGGACGACGACGACAAGTCCGCCGCATTGCGCGCGAGAGGACTGGCCGCTCTCAGAGCAGAAGCCGAAAGGCTGGGCCTCGACCCTGCAGCAGTCAGGCTCGTGGCAGTGCTGCGGGTGGGAGACAGGGTGGTGGCTTTGGGAGAGCTGCCCCCAGAAGCATAACCCCCACGCTGACCCAGACGATAGCAGAGTGCGCTTCGAGGCTGAGGTGCCCCGCTGGCGGGGGCAGCGGCGGGGGCTCCGCCCGCCATACTCTGCCCCTGACCGTGCCCTGAGGGCAAACAGACGTATACGTGGATGGAAAGCTGGTGGCCACAGTGCCTGCTGGGGAGCAGGAGGTGACGGTGGAAGTGCCCCCGGAACCCATCGCGTGAGCCTCCGCAATCGGAAGACTGGCGAGAGCAGAAGCTCCGATGTGGTCTTCCCTATCATTGTGGCGGTGGTGAT

ID	Sequence
A3B1C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA GCAAAGAAATCGAGAAGGAAGCCGAGAAAGCCTTCGAGGAGAAAAAATACGAAGAGGCTATCAAAGTGTACGAGGAGGCATTGAAGAAGGACCCA TCTAAGAAGCTGTACTACAAGGAAGGGATTGCCTTCGCATATTACGAACTGGGCAAAAAAGAGGAGGCAAGGAACTGTTTCATCGAGATCGCGGAA GAACTGGACTACAAGGACGACGACGACAAGTCCGCCGCATTGCGCGCGAGAGGACTGGCCGCTCTCAGAGCAGAAGCCGAAAGGCTGGGCCTCGA CCCTGCAGCAGTCAGGCTCGTGGCAGTGCTGCGGGTGGGAGACAGGGTGGTGGCTTTGGGAGAGCTGCCCCAGAAGCATAACCCACGCTGACCC AGACGATAGCAGAGTGCGCTTCGAGGCTGAGGTGCCCCGCTGGCGGGGGCAGCGGCGGGGGCTCCGGAGCCGCCGAGACAAGCCGCAGCCCTGG ACGCCGCCGCCGAAGCAGCCGAGGCTGCGGGAGACTTGGCCGCAGCTGCAGCAGCATATGCAGCCGCTGCAGCTCTCCTGGCAGCTCACGCCGCCA CTCATCCGAATGGCGCCAACCTTGGCCGCTTTGGCTGCTGAAGCCGCCGCTAGAGCCGCCGCCCTGGCTGCTCTGGCGGCCATCATTGTGGCGGTGGT GAT
A3B1C3	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA GCAAAGAAATCGAGAAGGAAGCCGAGAAAGCCTTCGAGGAGAAAAAATACGAAGAGGCTATCAAAGTGTACGAGGAGGCATTGAAGAAGGACCCA TCTAAGAAGCTGTACTACAAGGAAGGGATTGCCTTCGCATATTACGAACTGGGCAAAAAAGAGGAGGCAAGGAACTGTTTCATCGAGATCGCGGAA GAACTGGACTACAAGGACGACGACGACAAGTCCGCCGCATTGCGCGCGAGAGGACTGGCCGCTCTCAGAGCAGAAGCCGAAAGGCTGGGCCTCGA CCCTGCAGCAGTCAGGCTCGTGGCAGTGCTGCGGGTGGGAGACAGGGTGGTGGCTTTGGGAGAGCTGCCCCAGAAGCATAACCCACGCTGACCC AGACGATAGCAGAGTGCGCTTCGAGGCTGAGGTGCCCCGCTGGCGGGGGCAGCGGCGGGGGCTCCGGCTTTAGAGCAGCTGTCCACTGAGAGTCGG GGACCCGGTCGAAATTAGGCTGGACGACGGGACAGTGGTTCCAGGCAGGGTCGATCGCGTGGAGGGCGACGTGGTCTCCGTGGCCCCCTCTTGAGCC ACTCACCCCGGGCGTGACCGGGTGACCGTGAGGTTTCTGACCGAGCCTCCGCAGACCAGCGAGCCCAGACTGGTGGAGTTTCCCCCGGGTTCC CATCATTGTGGCGGTGGTGAT
A3B2C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA GCAAAGAAATCGAGAAGGAAGCCGAGAAAGCCTTCGAGGAGAAAAAATACGAAGAGGCTATCAAAGTGTACGAGGAGGCATTGAAGAAGGACCCA TCTAAGAAGCTGTACTACAAGGAAGGGATTGCCTTCGCATATTACGAACTGGGCAAAAAAGAGGAGGCAAGGAACTGTTTCATCGAGATCGCGGAA GAACTGGACTACAAGGACGACGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGACCCCTCCCAGATGGCACAGTGGTGACCGTAACTGTCTCTT GCACCGGACCCGGCGAGTACACGTGGGAGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGCACCCCTCCTTGCACCGTGACTTGACCTGTTC CCCCGACGACCCCGGCGGGGGCAGCGGCGGGGGCTCCGCCGCCATACTCTGCCCCTGACCGTGCTGAAGGCGAAACAGACGTATACGTGGATGG AAAGCTGGTGGCCACAGTGCCTGCTGGGGAGCAGGAGGTGACGGTGGAAGTGCCCCCGGAACCCATCGCGTGAGCCTCCGCAATCGGAAGACTG GCGAGAGCAGAAGCTCCGATGTGGTCTTCCCTATCATTGTGGCGGTGGTGAT

ID	Sequence
A3B2C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA GCAAAGAAATCGAGAAGGAAGCCGAGAAAGCCTTCGAGGAGAAAAAATACGAAGAGGCTATCAAAGTGTACGAGGAGGCATTGAAGAAGGACCCA TCTAAGAAGCTGTACTACAAGGAAGGGATTGCCTTCGCATATTACGAACTGGGCAAAAAAGAGGAGGCAAGGAAACTGTTTCATCGAGATCGCGGAA GAACTGGACTACAAGGACGACGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGCACCCCTCCAGATGGCACAGTGGTGACCGTAACTGTCTCTT GCACCGGACCCGGCGAGTACACGTGGGAGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGCACCCCTCCTTGCACCGTGACTTGACCTGTTC CCCCGACGACCCCGGCGGGGGCAGCGGCGGGGGCTCCGGAGCCGCCGCCAGACAAGCCGCAGCCCTGGACGCCGCCGCCGAAGCAGCCGAGGCT GCGGGAGACTTGCCGCAGCTGCAGCAGCATATGCAGCCGCTGCAGCTCTCCTGGCAGCTCACGCCGCCACTCATCCGAATGGCGCCAACCTTGGCCG CTTTGGCTGCTGAAGCCGCCGCTAGAGCCGCCGCCCTGGCTGCTCTGGCGGCCATCATTGTGGCGGTGGTGAT
A3B2C3	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA GCAAAGAAATCGAGAAGGAAGCCGAGAAAGCCTTCGAGGAGAAAAAATACGAAGAGGCTATCAAAGTGTACGAGGAGGCATTGAAGAAGGACCCA TCTAAGAAGCTGTACTACAAGGAAGGGATTGCCTTCGCATATTACGAACTGGGCAAAAAAGAGGAGGCAAGGAAACTGTTTCATCGAGATCGCGGAA GAACTGGACTACAAGGACGACGACGACAAGTCAGTAAAGGTCGTTACCTCTCAGTGCACCCCTCCAGATGGCACAGTGGTGACCGTAACTGTCTCTT GCACCGGACCCGGCGAGTACACGTGGGAGGTGAGCTTTTACGACGCCGACGATCCCAGCAATCGCACCCCTCCTTGCACCGTGACTTGACCTGTTC CCCCGACGACCCCGGCGGGGGCAGCGGCGGGGGCTCCGGCTTTAGAGCAGCTGTCCCACTGAGAGTCGGGGACCCGGTCGAAATTAGGCTGGACG ACGGGACAGTGGTTCCAGGCAGGGTCGATCGCGTGGAGGGCGACGTGGTCTCCGTGGCCCCCTCTTGAGCCACTACCCCGGGCGTGCACCGGGTGA CCGTGAGGTTCTTGACCGAGCCTCCGCAGACCAGCGAGCCCAGACTGGTGGAGTTTCCCCCGGGTTCCCATCATTGTGGCGGTGGTGAT
A3B3C1	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA GCAAAGAAATCGAGAAGGAAGCCGAGAAAGCCTTCGAGGAGAAAAAATACGAAGAGGCTATCAAAGTGTACGAGGAGGCATTGAAGAAGGACCCA TCTAAGAAGCTGTACTACAAGGAAGGGATTGCCTTCGCATATTACGAACTGGGCAAAAAAGAGGAGGCAAGGAAACTGTTTCATCGAGATCGCGGAA GAACTGGACTACAAGGACGACGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCGCAGAAGCCAGAGCTGCTTTGCTGGCCCAGGC AGCCGAACCTGGGCTGGACTCCGAGCGATCTCGAGTTGGAGGTGCGGCCAGTCCCCGAGGGAGTGCTGGCAATCGCACGCGCTACAGCCACCAGACC TGGCCAACCGCCCGTGAGATTGAGAGCCGAGGCCCTCCTGCGCCTGGGCGGGGGCAGCGGCGGGGGCTCCGCCCGCCATACTCTGCCCCTGACCGT GCCTGAAGGCGAAACAGACGTATACGTGGATGGAAAGCTGGTGGCCACAGTGCCTGCTGGGGAGCAGGAGGTGACGGTGGAAGTGCCCCCGGAA CCCATCGCGTGAGCCTCCGCAATCGGAAGACTGGCGAGAGCAGAAGCTCCGATGTGGTCTTCCCTATCATTGTGGCGGTGGTGAT

ID	Sequence
A3B3C2	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA GCAAAGAAATCGAGAAGGAAGCCGAGAAAGCCTTCGAGGAGAAAAAATACGAAGAGGCTATCAAAGTGTACGAGGAGGCATTGAAGAAGGACCCA TCTAAGAAGCTGTACTACAAGGAAGGGATTGCCTTCGCATATTACGAACTGGGCAAAAAAGAGGAGGCAAGGAACTGTTTCATCGAGATCGCGGAA GAACTGGACTACAAGGACGACGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCGCAGAAGCCAGAGCTGCTTTGCTGGCCCAGGC AGCCGAACTGGGCTGGACTCCGAGCGATCTCGAGTTGGAGGTGCGGCCAGTCCCCGAGGGAGTGCTGGCAATCGCACGCGCTACAGCCACCAGACC TGGCCAACCGCCCGTGAGATTGAGAGCCGAGGCCCTCCTGCGCCTGGGCGGGGGCAGCGGCGGGGGCTCCGGAGCCGCCGCCAGACAAGCCGCAG CCCTGGACGCCGCCGCCGAAGCAGCCGAGGCTGCGGGAGACTTGGCCGCAGCTGCAGCAGCATATGCAGCCGCTGCAGCTCTCCTGGCAGCTCACG CCGCCACTCATCCGAATGGCGCCAACTTGGCCGCTTTGGCTGCTGAAGCCGCCGCTAGAGCCGCCGCCCTGGCTGCTCTGGCGGCCATCATTGTGGC GGTGGTGAT
A3B3C3	AGCTGTGACCGGCGCCTACTCTAGAATGGCCCTCTGGATGAGGCTGCTGCCCCTGCTGGCATTGCTGGCCCTGTGGGGCCCTGATCCCGCTGCCGCAA GCAAAGAAATCGAGAAGGAAGCCGAGAAAGCCTTCGAGGAGAAAAAATACGAAGAGGCTATCAAAGTGTACGAGGAGGCATTGAAGAAGGACCCA TCTAAGAAGCTGTACTACAAGGAAGGGATTGCCTTCGCATATTACGAACTGGGCAAAAAAGAGGAGGCAAGGAACTGTTTCATCGAGATCGCGGAA GAACTGGACTACAAGGACGACGACGACAAGGCACCATCAGCCGAACAGTTGGCCGCCCTTGCCGCAGAAGCCAGAGCTGCTTTGCTGGCCCAGGC AGCCGAACTGGGCTGGACTCCGAGCGATCTCGAGTTGGAGGTGCGGCCAGTCCCCGAGGGAGTGCTGGCAATCGCACGCGCTACAGCCACCAGACC TGGCCAACCGCCCGTGAGATTGAGAGCCGAGGCCCTCCTGCGCCTGGGCGGGGGCAGCGGCGGGGGCTCCGGCTTTAGAGCAGCTGTCCCACTGAG AGTCGGGGACCCGGTCGAAATTAGGCTGGACGACGGGACAGTGGTTCCAGGCAGGGTCGATCGCGTGAGGGGCGACGTGGTCTCCGTGGCCCCCTCT TGAGCCACTCACCCCGGGCGTGACCCGGGTGACCGTGAGGTTCTTGACCGAGCCTCCGCAGACCAGCGAGCCCAGACTGGTGGAGTTTCCCCCCCG GGTTCATCATTGTGGCGGTGGTGAT

A total of 27 constructs were generated by combining 3 variants for each of the three segments. ID, identifier of the assembled sequence, where A, B, and C denote the three segments, and the numbers indicate the specific variant of each segment. Sequence, the full-length DNA sequence of the assembled construct (5' to 3').

Extended Data Table 3. Sequencing results of the 27 candidate sequences after the first round of screening.

Final Type	Total Counts
A2B3C3	2530
A2B3C2	1157
A3B2C2	1012

Final Type	Total Counts
A1B2C2	942
A1B3C3	919
A1B1C2	811
A3B1C2	782
A3B2C3	700
A1B1C1	657
A1B3C1	559
A2B1C3	553
A2B2C2	545
A1B1C3	472
A2B2C1	411
A2B1C1	396
A3B3C3	371
A3B3C1	317
A3B3C2	317
A3B1C1	297
A3B2C1	277
A1B2C3	270
A1B2C1	208
A2B3C1	150
A1B3C2	123
A2B1C2	118
A2B2C3	0
A3B1C3	0

Final Type, identifier of the final assembled construct, where A, B, and C denote the three segments and the numbers indicate the specific variant of each segment. Total Counts, the total number of sequencing reads obtained for each specific assembled type. Data are sorted in descending order of Total Counts.

Extended Data Table 4. Sequencing statistics of the enriched library after the first round of screening.

SumTotal Counts	Seq Counts	Perfect Percentage
14894	19606	75.9665%

Summary of next-generation sequencing (NGS) statistics for the enriched library. Sum Total Counts, the total number of sequencing reads corresponding to the target sequences. Seq Counts, the total number of valid sequencing reads obtained from the enriched library. Perfect Percentage, the percentage of correctly assembled sequences, calculated as (Sum Total Counts / Seq Counts) × 100%.

Extended Data Table 5. 90 repeated AlphaFold3 predictions of the A2B3C3 virtual complex

	ipTM	pTM	BSA (Å ²)	Hydrogen bond analysis		Hydrophobic pairing		Interface pLDDT	
				effective bonds	average energy	mimic with IL6	effective bonds	average energy	mimic with sIL6R
A2B3C3_1	0.62	0.66	1829.733	14	-19.71	17	14	75.7	49.4
A2B3C3_2	0.62	0.65	1877.327	11	-14.33	18	14	75.6	41.3
A2B3C3_3	0.63	0.67	1834.68	16	-19.48	17	21	75	45.4
A2B3C3_4	0.62	0.66	1573.99	14	-21.46	18	19	74.1	48.3
A2B3C3_5	0.63	0.68	1834.834	14	-17.01	18	23	75.3	45.7
A2B3C3_6	0.66	0.67	1999.601	14	-15.65	17	14	75.3	46.1
A2B3C3_7	0.59	0.65	1831.701	14	-19.08	18	8	74.4	39.6
A2B3C3_8	0.63	0.66	1684.835	14	-11.73	17	14	75.2	46.2
A2B3C3_9	0.65	0.66	1938.262	9	-8.65	18	16	74.7	43.4
A2B3C3_10	0.62	0.66	2433.042	19	-21.07	20	19	72.4	40

A2B3C3_11	0.63	0.67	1788.008	13	-15.53	18	21	75.6	45.8
A2B3C3_12	0.65	0.66	1628.776	11	-14.18	18	10	76.1	42
A2B3C3_13	0.64	0.66	2002.86	10	-11.51	19	6	75.3	39.5
A2B3C3_14	0.65	0.67	1871.37	11	-14.51	18	16	74.9	61.8
A2B3C3_15	0.64	0.68	1522.365	12	-14.31	18	5	76.1	56.7
A2B3C3_16	0.66	0.67	2190.491	10	-9.01	18	12	75.5	43.6
A2B3C3_17	0.63	0.65	1653.021	11	-17.26	18	18	74.5	34.1
A2B3C3_18	0.63	0.66	2222.7	10	-10.95	18	16	70.3	41
A2B3C3_19	0.62	0.66	1408.613	13	-13.29	18	15	75.1	47.5
A2B3C3_20	0.64	0.66	2224.081	17	-18.53	17	19	73.2	45.5
A2B3C3_21	0.63	0.66	1772.101	14	-19.6	18	10	76.3	41.3
A2B3C3_22	0.66	0.66	2309.83	13	-12.25	18	10	75.4	44.7
A2B3C3_23	0.62	0.67	1758.063	14	-13.73	18	26	75.4	42.8
A2B3C3_24	0.64	0.67	1989.224	9	-11.73	17	17	75	45.3
A2B3C3_25	0.64	0.67	1993.652	9	-10.44	18	18	75.2	52.6
A2B3C3_26	0.62	0.67	2037.689	19	-17.3	19	24	75	42.6
A2B3C3_27	0.63	0.65	1761.003	12	-8.27	17	21	75.1	46.1
A2B3C3_28	0.58	0.65	1808.27	15	-14.35	19	3	73.7	49.5
A2B3C3_29	0.65	0.66	2543.582	15	-14.26	19	26	70.2	43.2

A2B3C3_30	0.62	0.66	1588.402	16	-17.1	17	15	74.3	46.2
A2B3C3_31	0.62	0.66	1776.255	14	-16.2	19	18	76	57.6
A2B3C3_32	0.65	0.67	2008.63	15	-19.05	18	31	77.7	40.1
A2B3C3_33	0.64	0.67	1867.761	12	-16.3	18	9	74.5	56.4
A2B3C3_34	0.65	0.67	2151.016	10	-15.38	19	8	76.4	41.5
A2B3C3_35	0.64	0.66	2428.659	14	-18.06	19	16	66	43.2
A2B3C3_36	0.64	0.65	1650.311	16	-17.95	19	16	75.5	41.4
A2B3C3_37	0.63	0.66	2281.445	13	-15.02	18	11	74.8	42.3
A2B3C3_38	0.65	0.66	1658.327	13	-15.29	18	11	74.2	51.4
A2B3C3_39	0.59	0.67	2358.384	20	-26.31	18	31	65.3	43.2
A2B3C3_40	0.63	0.67	1769.669	15	-18.11	17	18	75.2	44.1
A2B3C3_41	0.62	0.71	1828.411	17	-21.04	18	26	75.5	70.2
A2B3C3_42	0.65	0.67	2064.925	14	-17.97	18	15	75.4	41.6
A2B3C3_43	0.63	0.67	1555.423	18	-24.03	18	23	74.6	44.1
A2B3C3_44	0.63	0.67	2247.854	15	-17.01	18	12	74.6	42.6
A2B3C3_45	0.67	0.71	1771.375	16	-24.18	19	15	76.5	67.6
A2B3C3_46	0.64	0.66	2356.609	10	-15.89	16	13	74	42.7
A2B3C3_47	0.64	0.66	1506.355	18	-25.43	18	18	75.6	54.4
A2B3C3_48	0.68	0.68	1626.115	19	-16.76	18	27	77	51.5

A2B3C3_49	0.64	0.67	1938.033	13	-19.6	16	22	76	43.2
A2B3C3_50	0.65	0.67	1373.333	9	-11.08	18	13	74.4	45.3
A2B3C3_51	0.64	0.68	9314.033	13	-14.3	18	21	75.1	46.8
A2B3C3_52	0.63	0.66	1601.409	10	-14.17	18	7	75.7	51.2
A2B3C3_53	0.63	0.67	2201.611	12	-18.29	17	22	75.2	41.6
A2B3C3_54	0.62	0.66	1827.47	18	-20.76	18	34	76	46.5
A2B3C3_55	0.62	0.66	1768.302	18	-21.54	19	18	75.5	39.7
A2B3C3_56	0.6	0.66	1797.276	14	-18.31	18	13	74.8	55.1
A2B3C3_57	0.62	0.66	1964.762	11	-16.37	18	10	74.6	38.6
A2B3C3_58	0.63	0.67	1642.997	13	-14.36	19	25	75.5	39.5
A2B3C3_59	0.64	0.67	1722.606	8	-8.09	19	13	76.1	68.8
A2B3C3_60	0.61	0.67	1606.459	10	-10.4	18	20	74.8	42
A2B3C3_61	0.63	0.67	1791.736	12	-9.37	17	32	66.66	40.81
A2B3C3_62	0.64	0.66	1884.295	17	-21.53	17	15	75.21	63.42
A2B3C3_63	0.63	0.67	1809.749	14	-12.63	18	15	71.47	55.07
A2B3C3_64	0.65	0.67	1902.02	11	-21.67	19	18	67.74	50.23
A2B3C3_65	0.61	0.66	1999.601	11	-10.8	18	32	76.16	62.91
A2B3C3_66	0.62	0.65	1798.491	9	-23.14	18	11	68.63	53.75
A2B3C3_67	0.62	0.66	1798.493	15	-11.9	19	25	68.24	49.2

A2B3C3_68	0.65	0.67	1999.601	11	-9.23	19	9	69.92	34.79
A2B3C3_69	0.65	0.66	1916.078	17	-18.13	18	24	70.09	58.78
A2B3C3_70	0.64	0.66	1770.864	15	-19.3	16	9	68.19	51.72
A2B3C3_71	0.62	0.67	1889.678	16	-19.8	20	24	67.48	42.06
A2B3C3_72	0.62	0.66	1771.575	17	-22.79	19	15	70.24	53.97
A2B3C3_73	0.62	0.67	1771.304	18	-24.48	17	30	75.74	52.97
A2B3C3_74	0.65	0.66	1854.387	18	-17.52	17	10	75.06	51.7
A2B3C3_75	0.63	0.66	1601.36	16	-15.13	20	20	73.25	43.6
A2B3C3_76	0.63	0.67	1623.474	15	-8.98	19	13	71.45	53.14
A2B3C3_77	0.64	0.66	1759.968	14	-19.96	20	18	71.17	69.64
A2B3C3_78	0.64	0.68	1707.074	12	-21.27	20	8	71.18	56.61
A2B3C3_79	0.65	0.66	1862.874	8	-18.17	20	7	77	56.83
A2B3C3_80	0.64	0.66	1719.378	15	-8.86	19	23	74.45	66.95
A2B3C3_81	0.62	0.66	1660.175	16	-23.04	19	12	65.82	68.85
A2B3C3_82	0.64	0.66	1998.049	8	-23.71	16	12	74.7	55.22
A2B3C3_83	0.64	0.66	1799.475	19	-24.79	20	32	71.67	40.05
A2B3C3_84	0.65	0.68	1833.909	15	-16.34	19	29	67.51	48.24
A2B3C3_85	0.64	0.67	1658.714	18	-20.66	16	3	70.16	68.2
A2B3C3_86	0.63	0.66	1762.07	14	-21.16	18	31	76.95	46.5

A2B3C3_87	0.65	0.67	1839.003	11	-9.53	18	33	67.94	59.55
A2B3C3_88	0.61	0.67	1690.853	16	-18.41	19	25	75.38	50.81
A2B3C3_89	0.64	0.66	1870.088	13	-18.68	18	5	70.31	48.06
A2B3C3_90	0.65	0.68	1755.465	13	-8.34	19	15	72.58	53.18

A total of 90 independent AlphaFold3 predictions were performed and shown. ipTM, interface predicted TM-score (0–1, higher indicates higher confidence). pTM, predicted TM-score. BSA ($\text{\AA}^2$), buried surface area. For Hydrogen bond analysis and Hydrophobic pairing, effective bonds denote the count of interfacial hydrogen bonds or hydrophobic contacts, and average energy represents the average interaction energy (kcal/mol). mimic with IL6 and mimic with sIL6R α refer to the evaluation metrics indicating how well the predicted binder interface mimics the native binding mode of IL-6 and sIL-6R α , respectively. For Interface pLDDT, average is the mean predicted local distance difference test score across interfacial residues (0–100, higher is better), and mimic with sIL6R α indicates the interface pLDDT when the binder adopts the sIL-6R α binding mode. These static structural descriptors should not be interpreted as binding free energies or molecular dynamics trajectory averages.

Extended Data Table 6. 90 repeated AlphaFold3 predictions of the A2B3C2 virtual complex

	ipTM	pTM	BSA (Å ²)	Hydrogen bond analysis		Hydrophobic pairing		Interface pLDDT	
				effective bonds	average energy	mimic with IL6	effective bonds	average energy	mimic with IL6
A2B3C2_1	0.62	0.71	1473.948	19	-25.63	16	10	72.4	70.4
A2B3C2_2	0.6	0.7	1402.993	18	-26.41	18	10	72.3	68.6
A2B3C2_3	0.62	0.71	1367.934	17	-27.36	20	9	71.4	71.1
A2B3C2_4	0.6	0.69	1720.724	15	-18.22	18	11	71.6	68
A2B3C2_5	0.62	0.71	1771.421	16	-19.57	19	10	71.2	71.9
A2B3C2_6	0.6	0.69	1354.058	18	-25.7	19	9	72.9	69.4
A2B3C2_7	0.62	0.71	1431.637	20	-29.53	18	11	71.4	71.6
A2B3C2_8	0.62	0.71	1581.135	17	-25.74	18	11	71.4	70.2
A2B3C2_9	0.6	0.69	1623.096	16	-19.82	20	11	73.9	68
A2B3C2_10	0.6	0.69	1442.243	18	-23.09	18	11	73.4	68.9
A2B3C2_11	0.62	0.71	1295.91	17	-24.12	17	10	71.3	70.4
A2B3C2_12	0.63	0.72	1594.086	22	-27.53	19	10	70.9	70.7
A2B3C2_13	0.6	0.69	1580.736	18	-23.75	18	11	70.7	69
A2B3C2_14	0.64	0.72	1587.935	20	-27.55	19	9	72	71.2
A2B3C2_15	0.6	0.69	1500.211	17	-22.5	20	11	72.8	67.4
A2B3C2_16	0.6	0.69	1494.192	19	-30.04	18	13	73.8	68.2

A2B3C2_17	0.62	0.71	1554.415	16	-23.03	19	10	72.3	71.6
A2B3C2_18	0.6	0.69	1354.87	22	-33.49	21	10	71.7	70.6
A2B3C2_19	0.61	0.7	1666.273	16	-24.46	18	11	71.4	70.6
A2B3C2_20	0.6	0.69	1525.943	17	-18.98	18	9	72.9	66.7
A2B3C2_21	0.64	0.72	1678.048	19	-24.81	17	11	71.7	71.7
A2B3C2_22	0.64	0.72	1394.159	16	-22.44	18	10	73.5	73.6
A2B3C2_23	0.62	0.71	1521.908	19	-27.53	20	12	71.6	70
A2B3C2_24	0.63	0.71	1777.792	17	-23.75	19	9	71.2	69.8
A2B3C2_25	0.6	0.69	1652.654	19	-29.98	19	11	71.8	69.5
A2B3C2_26	0.6	0.69	1596.079	18	-26.15	20	12	71.5	68.7
A2B3C2_27	0.6	0.69	1561.741	15	-20.11	17	13	70.5	66
A2B3C2_28	0.62	0.71	1391.571	21	-33.92	17	10	70.6	72.6
A2B3C2_29	0.6	0.69	1364.3	21	-31.15	20	10	72.3	70.1
A2B3C2_30	0.62	0.71	1646.259	20	-30.44	19	10	70.5	69.9
A2B3C2_31	0.6	0.7	1524.079	16	-25.45	18	12	73.1	70.4
A2B3C2_32	0.63	0.71	1725.014	16	-25.93	19	14	71.8	70
A2B3C2_33	0.6	0.69	1401.853	17	-25.45	17	8	72.7	70.1
A2B3C2_34	0.64	0.72	1423.633	15	-20.43	19	9	71.5	71.5
A2B3C2_35	0.64	0.72	1710.533	18	-23.87	21	11	71.9	71.6

A2B3C2_36	0.6	0.69	1393.989	20	-31.56	20	15	69.9	70
A2B3C2_37	0.6	0.69	1439.228	17	-26.01	20	11	72.4	67.9
A2B3C2_38	0.6	0.69	1757.173	17	-23.9	19	10	72.2	67.6
A2B3C2_39	0.62	0.7	1506.595	17	-25.02	19	11	74.7	71.3
A2B3C2_40	0.6	0.69	1612.364	19	-26.2	18	11	72.4	69
A2B3C2_41	0.59	0.69	1485.701	20	-33.98	18	10	72.7	68.4
A2B3C2_42	0.61	0.7	1463.443	20	-25.09	20	11	71.7	68.6
A2B3C2_43	0.63	0.72	1518.177	11	-14.99	19	11	72.1	72.2
A2B3C2_44	0.59	0.68	1507.327	13	-15.97	19	11	72.7	68.9
A2B3C2_45	0.6	0.69	1359.571	16	-22.96	17	12	72.4	69.5
A2B3C2_46	0.6	0.69	1438.948	20	-28.27	19	13	72.1	68.6
A2B3C2_47	0.6	0.69	12225.364	19	-26.62	18	11	72.9	69.6
A2B3C2_48	0.6	0.69	1473.016	15	-22.66	19	12	71.7	66.7
A2B3C2_49	0.62	0.71	1769.41	17	-24.97	18	12	71.7	69.9
A2B3C2_50	0.59	0.69	1360.698	18	-22.35	19	10	72.4	67.9
A2B3C2_51	0.63	0.72	1440.986	16	-25.01	18	10	71.2	72.1
A2B3C2_52	0.63	0.72	1444.82	20	-27.08	20	11	72.7	70.4
A2B3C2_53	0.64	0.72	1629.793	21	-25.89	19	15	72.6	70.6
A2B3C2_54	0.58	0.68	1534.85	19	-22.54	19	11	72.8	69

A2B3C2_55	0.59	0.68	1532.948	21	-26.93	20	10	72.9	70.6
A2B3C2_56	0.59	0.68	747.409	19	-28.36	19	10	72	68.5
A2B3C2_57	0.6	0.69	1297.992	21	-28.79	21	11	72.7	71.2
A2B3C2_58	0.6	0.7	1687.079	15	-21.82	20	10	72	70
A2B3C2_59	0.61	0.7	1592.092	22	-29.56	20	10	72.5	67.7
A2B3C2_60	0.59	0.69	1405.245	19	-29.09	19	10	71.7	69
A2B3C2_61	0.61	0.7	1382.779	22	-15.21	20	11	74.6	70.5
A2B3C2_62	0.6	0.7	1305.295	19	-25.06	16	14	72.9	73.1
A2B3C2_63	0.62	0.71	1519.127	16	-18.34	21	14	71.2	70.1
A2B3C2_64	0.63	0.72	1283.134	17	-27.98	21	11	72	70.6
A2B3C2_65	0.6	0.69	1645.243	16	-33.02	21	10	73.1	67.7
A2B3C2_66	0.6	0.69	1498.399	18	-19.85	19	11	72.9	71.4
A2B3C2_67	0.61	0.7	1529.829	13	-28.98	21	11	73.9	70
A2B3C2_68	0.6	0.69	1302.069	14	-15.42	20	10	72.5	69.8
A2B3C2_69	0.64	0.72	1727.768	20	-24.37	19	10	70.6	70.3
A2B3C2_70	0.61	0.7	1530.766	22	-21.34	18	8	74.3	67
A2B3C2_71	0.6	0.69	1749.75	12	-15.11	21	9	71.9	67.1
A2B3C2_72	0.6	0.69	1705.893	11	-27.9	21	11	71.3	73.1
A2B3C2_73	0.6	0.69	1647.667	14	-19	18	14	72.4	73.3

A2B3C2_74	0.63	0.72	1426.384	17	-26.53	20	10	71.2	70.9
A2B3C2_75	0.61	0.7	1348.308	19	-29.9	16	10	74.3	73.3
A2B3C2_76	0.59	0.69	1726.666	18	-15.92	17	11	73.4	68.7
A2B3C2_77	0.59	0.69	1708.957	12	-27.62	16	15	72.3	66.3
A2B3C2_78	0.61	0.7	1375.462	15	-24.45	21	8	72.3	70.2
A2B3C2_79	0.64	0.73	1545.285	18	-19.89	20	12	72.2	72.3
A2B3C2_80	0.59	0.69	1600.62	19	-20.3	19	8	73	71.6
A2B3C2_81	0.59	0.68	1305.781	13	-26.92	20	11	72	70.9
A2B3C2_82	0.62	0.7	1352.164	11	-30.97	20	11	72.5	66.4
A2B3C2_83	0.63	0.72	1612.757	16	-19.9	19	9	71.2	71.4
A2B3C2_84	0.6	0.7	1448.362	14	-32.95	18	15	74.6	67.8
A2B3C2_85	0.61	0.7	1372.005	20	-21.62	18	10	74.4	71.4
A2B3C2_86	0.64	0.72	1623.726	19	-22.46	20	10	72.6	71.4
A2B3C2_87	0.63	0.72	1607.436	16	-23.84	19	12	70.7	71.3
A2B3C2_88	0.59	0.68	1661.649	16	-33.22	20	9	73.9	66.4
A2B3C2_89	0.6	0.69	1690.222	21	-29.65	20	15	70.5	72.4
A2B3C2_90	0.62	0.7	1461.978	17	-25.29	18	13	72.6	73.5

A total of 90 independent AlphaFold3 predictions were performed and shown. ipTM, interface predicted TM-score (0–1, higher indicates higher confidence). pTM, predicted TM-score. BSA (Å²), buried surface area in square angstroms. For Hydrogen bond analysis and Hydrophobic pairing, effective bonds denote the count of interfacial hydrogen bonds or hydrophobic contacts, and average energy represents the average interaction energy (in kcal/mol). mimic with IL-6 denotes the number of interfacial bonds [or structural overlap metrics] that recapitulate the native binding interfaces of IL-6. For Interface pLDDT, average is the mean predicted local distance difference test score across interfacial

residues (0–100, higher is better), and mimic with IL-6 indicates the interface pLDDT when the binder adopts the IL-6 binding mode.

Extended Data Table 7. AlphaFold3 predictions of A2B3C3 (ipTM > 0.66) and A2B3C2 (ipTM > 0.68) used for virtual complex model construction

	BSA (Å ²)	Hydrogen Bond Analysis		hydrophobic pairing	
		Effective Bonds	Average Energy	mimic with IL6	mimic with sIL6R
A2B3C3	1801.79	13	-12.19	18	23
A2B3C2	1759.434	13	-17.82	19	10
A3B2C2	1301.479	28	-40.46	-	41

Summary of AlphaFold3 predictions for the selected candidates used to construct the virtual complex models. BSA (Å²), buried surface area in square angstroms. For Hydrogen Bond Analysis, Effective Bonds denote the number of interfacial hydrogen bonds, and Average Energy represents the average interaction energy (in kcal/mol). For hydrophobic pairing, mimic with IL6 and mimic with sIL6R denote the number of interfacial hydrophobic contacts that recapitulate the native binding interfaces of IL-6 and sIL-6R, respectively. A dash (-) indicates that the corresponding metric was not detected for the A3B2C2 complex.

Extended Data Table 8. PIA-derived interface characteristics of the analyzed protein–protein complexes.

System	Interfa cial Residue Pairs	Hydroge n Bonds (HBs)	Salt Bridg es	van der Waals (vdW) Clashes	Mean Positive Shape Complementar ity (SC)	Mean Residue (%)	Ligand Burial
candidate1 (A2B3C3)	68	17	5	0	0.628	51.4	
candidate3 (A3B2C2)	43	3	2	0	0.701	60.1	
candidate2 (A2B3C2)	68	13	5	0	0.574	54.4	
Reference (GP130)	102	9	2	2	0.715	56.1	
candidate1– sIL-6Rα	72	8	3	0	0.644	59.2	
candidate1– IL-6	53	17	2	0	0.608	58.4	

Interface statistics were calculated from one representative structure for each system. Interfacial residue pairs denote cross-interface residue–residue contacts identified by PIA. HBs, salt bridges, and van der Waals (vdW) clashes indicate the corresponding interaction counts. Mean positive shape complementarity (SC) and mean ligand-residue burial were reported as calculated by PIA. The candidate1–sIL-6Rα and candidate1–IL-6 complexes were generated by independent docking and were not obtained by deleting one

chain from the complete candidate 1 complex. These static structural descriptors should not be interpreted as binding free energies or molecular-dynamics trajectory averages.

Extended Table 9. 90 repeated AlphaFold3 predictions of the S1 virtual complex

	ipTM	pTM	BSA (Å ²)	Hydrogen bond analysis		Hydrophobic pairing	
				effective bonds	average energy	mimic with IL6	effective bonds
S1_1	0.51	0.63	2349.764	18	-26.34	17	12
S1_2	0.51	0.63	1360.39	11	-16.81	15	13
S1_3	0.53	0.64	2024.243	9	-10.05	14	13
S1_4	0.51	0.63	2032.243	14	-16.34	16	14
S1_5	0.53	0.64	2063.58	11	-16.78	16	13
S1_6	0.51	0.63	1909.652	8	-11.11	16	11
S1_7	0.52	0.63	2143.567	17	-17.12	16	22
S1_8	0.51	0.63	1878.599	14	-18.28	16	26
S1_9	0.52	0.63	2120.488	11	-10.39	30	11
S1_10	0.52	0.63	2316.62	10	-13.42	17	11
S1_11	0.52	0.64	1914.565	13	-9.45	15	14
S1_12	0.51	0.63	1924.331	14	-21.45	15	10
S1_13	0.51	0.63	1893.104	15	-24.51	16	19
S1_14	0.52	0.63	2778.603	15	-18.76	31	14
S1_15	0.53	0.64	1744.917	15	-18.56	15	9
S1_16	0.51	0.63	1849.511	14	-14.88	16	9

S1_17	0.5	0.62	1835.762	11	-10.75	17	8
S1_18	0.53	0.64	1671.753	11	-17.97	15	9
S1_19	0.51	0.62	1706.979	8	-9.82	16	26
S1_20	0.5	0.62	1744.795	20	-18.83	15	23
S1_21	0.5	0.62	2085.944	12	-18.02	20	11
S1_22	0.52	0.63	1583.452	18	-20.56	17	26
S1_23	0.51	0.63	1841.118	13	-10.3	14	25
S1_24	0.52	0.64	1863.401	18	-22.93	23	26
S1_25	0.52	0.63	2123.418	14	-13.25	31	26
S1_26	0.51	0.63	1936.263	15	-22.12	30	14
S1_27	0.52	0.64	1967.832	17	-12.19	28	19
S1_28	0.52	0.64	2695.306	20	-18.94	25	25
S1_29	0.51	0.63	1369.399	15	-10.44	24	11
S1_30	0.51	0.63	1850.636	9	-15.19	19	19
S1_31	0.51	0.63	1617.287	18	-25.89	17	13
S1_32	0.52	0.63	2130.463	12	-24.04	23	13
S1_33	0.52	0.63	2457.421	15	-11.18	18	18
S1_34	0.51	0.63	1834.801	10	-21.48	24	17
S1_35	0.51	0.63	1593.516	8	-22.82	19	12

S1_36	0.51	0.63	2222.373	20	-13.94	19	18
S1_37	0.52	0.64	1885.885	9	-11.36	25	18
S1_38	0.51	0.63	1817.928	19	-24.61	22	16
S1_39	0.56	0.65	2146.613	15	-22.3	15	19
S1_40	0.52	0.64	2456.064	18	-22.61	31	16
S1_41	0.5	0.62	2234.849	13	-17.07	21	24
S1_42	0.51	0.63	1970.99	16	-14.64	16	12
S1_43	0.51	0.63	2047.242	15	-21.25	28	18
S1_44	0.51	0.63	1984.277	9	-25.95	31	8
S1_45	0.53	0.64	1912.466	19	-10.84	14	15
S1_46	0.53	0.63	1899.164	20	-17.35	22	21
S1_47	0.51	0.63	1851.335	16	-16.17	29	8
S1_48	0.5	0.64	1670.124	9	-12.36	31	23
S1_49	0.52	0.63	1421.924	15	-10.1	27	8
S1_50	0.51	0.63	1606.133	10	-12.08	25	17
S1_51	0.52	0.63	2238.229	14	-18.31	14	21
S1_52	0.52	0.64	1807.015	14	-15.07	29	21
S1_53	0.53	0.63	1683.641	13	-24.42	22	25
S1_54	0.52	0.64	1475.981	17	-17.48	16	9

S1_55	0.52	0.62	1651.728	19	-13.06	21	9
S1_56	0.51	0.64	2026.674	15	-13.5	20	26
S1_57	0.52	0.64	1872.049	8	-12.07	26	8
S1_58	0.5	0.64	1641.598	13	-15.32	31	24
S1_59	0.51	0.62	1411.808	11	-20.05	27	19
S1_60	0.52	0.64	1501.604	9	-25.99	21	19
S1_61	0.51	0.64	1567.534	14	-10.49	25	11
S1_62	0.51	0.62	1446.498	17	-11.63	23	23
S1_63	0.52	0.63	1564.898	10	-22.74	23	17
S1_64	0.53	0.63	2099.609	16	-25.91	27	19
S1_65	0.52	0.64	2005.774	16	-10.96	28	18
S1_66	0.5	0.62	2007.084	12	-22.56	30	9
S1_67	0.53	0.63	1616.755	17	-14.36	31	26
S1_68	0.52	0.63	1788.079	9	-24.02	28	8
S1_69	0.53	0.63	1434.374	20	-11.98	30	26
S1_70	0.51	0.64	1974.08	8	-17.15	19	20
S1_71	0.52	0.62	1810.24	20	-10.32	28	24
S1_72	0.52	0.63	1773.291	10	-25.81	25	13
S1_73	0.51	0.63	2034.549	9	-12.19	14	12

S1_74	0.52	0.63	1394.483	14	-14.88	17	19
S1_75	0.51	0.63	1521.69	20	-24.34	24	24
S1_76	0.5	0.63	1453.746	16	-20.79	29	22
S1_77	0.52	0.64	1860.97	16	-20.83	23	12
S1_78	0.51	0.62	1622.152	12	-21.13	18	9
S1_79	0.52	0.62	1465.983	10	-18.31	31	12
S1_80	0.5	0.63	1716.267	11	-17.67	31	24
S1_81	0.51	0.63	1557.368	15	-17.17	14	15
S1_82	0.52	0.64	1835.905	8	-21.39	22	22
S1_83	0.5	0.62	1568.222	10	-13.22	25	13
S1_84	0.51	0.62	1748.81	18	-11.15	27	21
S1_85	0.51	0.63	1654.912	19	-14.27	31	14
S1_86	0.51	0.63	1532.144	13	-16.06	25	8
S1_87	0.52	0.63	1636.28	18	-21.98	31	19
S1_88	0.51	0.62	1545.976	11	-19.53	15	20
S1_89	0.51	0.63	1701.117	19	-26.04	14	23
S1_90	0.51	0.63	1594.928	16	-23.28	28	8

A total of 90 independent AlphaFold3 predictions were performed shown. ipTM, interface predicted TM-score (0–1, higher indicates higher confidence). pTM, predicted TM-score. BSA (Å²), buried surface area in square angstroms. For Hydrogen bond analysis, effective bonds denote the number of interfacial hydrogen bonds, and average energy represents the average interaction energy (in kcal/mol). mimic with IL6 denotes the number of interfacial hydrogen bonds that recapitulate the native binding interfaces of IL-6. For Hydrophobic pairing, effective bonds indicate the number of interfacial hydrophobic contacts.

Extended Table 10. 90 repeated AlphaFold3 predictions of the S2 virtual complex

	ipTM	pTM	BSA (Å ²)	Hydrogen bond analysis		Hydrophobic pairing	
				effective bonds	average energy	mimic with IL6	effective bonds
S2_1	0.52	0.63	1695.32	13	-20.24	18	3
S2_2	0.56	0.65	1905.99	12	-15.5	18	9
S2_3	0.52	0.64	1845.903	15	-14.56	17	11
S2_4	0.54	0.65	1288.401	15	-16.36	17	16
S2_5	0.52	0.63	2077.535	9	-9.72	17	17
S2_6	0.57	0.64	1782.901	14	-17.4	18	9
S2_7	0.53	0.64	1653.865	7	-10.81	19	2
S2_8	0.59	0.65	2146.193	10	-17.02	19	20
S2_9	0.55	0.63	1327.702	10	-15.01	19	3
S2_10	0.55	0.64	1545.423	11	-15.9	19	7
S2_11	0.53	0.64	2075.655	11	-20.21	17	8
S2_12	0.68	0.67	1570.663	9	-14.09	17	17
S2_13	0.54	0.64	1666.834	9	-19.07	19	7
S2_14	0.61	0.65	1981.477	7	-9.96	19	12
S2_15	0.53	0.64	1713.967	12	-19.55	19	18
S2_16	0.56	0.65	1337.438	13	-12.84	18	3

S2_17	0.55	0.64	2051.938	14	-17.14	17	18
S2_18	0.57	0.64	2016.807	9	-13.48	17	12
S2_19	0.57	0.64	1393.555	12	-16.83	19	11
S2_20	0.54	0.63	1608.409	8	-17.25	19	13
S2_21	0.65	0.63	1862.986	9	-19.61	18	5
S2_22	0.69	0.65	1743.842	11	-14.24	19	2
S2_23	0.59	0.66	1748.187	14	-11.21	17	9
S2_24	0.64	0.63	1618.889	14	-12.62	17	2
S2_25	0.68	0.63	1877.445	15	-15.5	18	14
S2_26	0.54	0.63	1323.088	8	-15.42	17	13
S2_27	0.54	0.63	1608.409	8	-16.35	17	10
S2_28	0.56	0.66	1802.612	15	-19.13	17	14
S2_29	0.52	0.63	1646.468	12	-11.83	18	17
S2_30	0.62	0.64	1919.159	10	-13.91	19	15
S2_31	0.59	0.63	1988.027	12	-11.91	19	7
S2_32	0.65	0.64	1676.653	12	-10.02	18	12
S2_33	0.56	0.64	1959.989	9	-10.9	18	2
S2_34	0.53	0.63	1327.061	15	-12.66	18	9
S2_35	0.71	0.63	1860.51	9	-12.13	17	3

S2_36	0.56	0.66	1991.81	7	-14.04	17	19
S2_37	0.56	0.63	1494.077	14	-11.36	19	4
S2_38	0.56	0.66	1749.186	12	-16.52	18	20
S2_39	0.52	0.65	1803.109	9	-18.55	18	15
S2_40	0.6	0.64	1329.033	13	-10.81	18	3
S2_41	0.53	0.64	1849.439	10	-14.5	19	16
S2_42	0.71	0.67	1590.004	10	-13.66	19	16
S2_43	0.54	0.64	1535.351	7	-16.25	19	3
S2_44	0.63	0.63	1424.219	7	-14.36	18	9
S2_45	0.55	0.65	1442.057	11	-19.85	18	4
S2_46	0.76	0.68	1659.077	13	-15.23	19	9
S2_47	0.71	0.67	2016.393	11	-14.45	18	2
S2_48	0.56	0.64	1657.577	9	-18.21	17	5
S2_49	0.58	0.65	1313.481	15	-19.39	17	6
S2_50	0.57	0.64	1505.654	15	-17.11	18	11
S2_51	0.56	0.64	1959.426	15	-14.92	18	6
S2_52	0.53	0.64	1817.85	15	-18.38	17	7
S2_53	0.52	0.64	1882.13	7	-13.37	17	9
S2_54	0.55	0.65	1499.398	8	-10.21	19	4

S2_55	0.55	0.64	1959.964	12	-15	18	2
S2_56	0.69	0.67	1477.338	7	-12.07	17	20
S2_57	0.57	0.65	1958.697	9	-19.44	18	18
S2_58	0.55	0.64	1487.825	12	-15.88	17	9
S2_59	0.57	0.65	1690.861	7	-17.06	17	14
S2_60	0.54	0.64	1854.632	13	-15.83	19	19
S2_61	0.52	0.64	1966.438	12	-18.49	18	18
S2_62	0.51	0.63	1781.512	12	-16.41	17	14
S2_63	0.57	0.65	1418.472	13	-17.18	17	11
S2_64	0.56	0.65	1467.64	13	-15.26	17	11
S2_65	0.57	0.64	1670.308	15	-19.15	19	14
S2_66	0.55	0.65	1397.543	9	-9.93	18	8
S2_67	0.77	0.68	2235.785	14	-18.25	19	8
S2_68	0.53	0.64	1574.226	11	-19.87	18	14
S2_69	0.55	0.65	1633.926	12	-13.82	18	9
S2_70	0.57	0.65	1763.877	7	-18.45	18	19
S2_71	0.55	0.65	1503.103	14	-10.03	18	17
S2_72	0.53	0.65	1311.414	12	-15.65	17	3
S2_73	0.55	0.64	1865.513	8	-16.22	19	16

S2_74	0.53	0.64	1723.215	11	-15.83	19	8
S2_75	0.56	0.64	1975.551	12	-15.13	17	12
S2_76	0.59	0.65	2052.199	15	-19.28	17	14
S2_77	0.53	0.64	2131.635	12	-11.23	18	9
S2_78	0.56	0.64	1409.411	12	-12.95	19	2
S2_79	0.52	0.64	1953.141	14	-13.97	19	20
S2_80	0.53	0.64	1722.173	12	-13.46	17	5
S2_81	0.56	0.64	1740.245	10	-16.82	18	5
S2_82	0.55	0.64	1571.556	10	-18.83	17	17
S2_83	0.53	0.64	1581.117	15	-12.2	19	20
S2_84	0.54	0.64	2102.818	15	-18.46	18	20
S2_85	0.54	0.65	1878.91	11	-18.22	17	4
S2_86	0.54	0.64	1474.2	12	-12.96	17	6
S2_87	0.54	0.64	2141.376	15	-14.59	17	6
S2_88	0.57	0.64	1732.069	10	-11.29	19	9
S2_89	0.52	0.63	2140.28	10	-18.51	17	11
S2_90	0.55	0.64	2052.395	15	-16.85	17	6

A total of 90 independent AlphaFold3 predictions were performed and shown. ipTM, interface predicted TM-score (0–1, higher indicates higher confidence). pTM, predicted TM-score. BSA (Å²), buried surface area in square angstroms. For Hydrogen bond analysis, effective bonds denote the number of interfacial hydrogen bonds, and average energy represents the average interaction energy (in kcal/mol). mimic with IL6 denotes the number of interfacial hydrogen bonds that recapitulate the native binding interfaces of IL-6. For Hydrophobic pairing, effective bonds indicate the number of interfacial hydrophobic contacts.

Extended Table 11. 90 repeated AlphaFold3 predictions of the S3 virtual complex

	ipTM	pTM	BSA (Å ²)	Hydrogen bond analysis		Hydrophobic pairing	
				effective bonds	average energy	mimic with IL6	effective bonds
S3_1	0.56	0.65	2364.587	11	-14.33	17	18
S3_2	0.52	0.64	1530.069	8	-13.37	17	12
S3_3	0.53	0.64	1586.203	12	-17.36	16	10
S3_4	0.56	0.65	2029.343	19	-24.93	17	16
S3_5	0.53	0.65	2024.159	15	-20.29	16	11
S3_6	0.52	0.64	2305.656	13	-15.36	15	24
S3_7	0.55	0.65	1646.323	11	-12.43	16	7
S3_8	0.57	0.65	1423.955	11	-13.38	16	4
S3_9	0.56	0.65	1659.785	11	-12.66	16	9
S3_10	0.52	0.64	1704.409	12	-16.73	17	15
S3_11	0.55	0.64	1432.347	18	-20.63	15	11
S3_12	0.54	0.65	1464.516	9	-22.41	17	14
S3_13	0.54	0.65	1550.833	13	-14.46	15	15
S3_14	0.53	0.64	1919.658	8	-24.35	16	10
S3_15	0.53	0.64	2266.437	8	-24.73	17	6
S3_16	0.57	0.65	1692.803	11	-24.21	17	23

S3_17	0.54	0.65	1632.586	13	-20.77	17	23
S3_18	0.53	0.64	1830.883	8	-22.38	15	5
S3_19	0.53	0.65	1431.469	8	-18.64	15	15
S3_20	0.53	0.64	1726.449	18	-17.93	17	22
S3_21	0.55	0.64	1974.366	11	-24.43	16	6
S3_22	0.57	0.65	1443.184	13	-24.08	16	23
S3_23	0.55	0.65	1858.176	12	-24.38	16	13
S3_24	0.52	0.64	1547.939	12	-15.59	16	13
S3_25	0.56	0.65	1979.198	15	-17.57	17	16
S3_26	0.53	0.64	1455.148	19	-24.46	17	5
S3_27	0.53	0.64	1902.916	13	-13.68	17	14
S3_28	0.56	0.65	1636.961	17	-18.59	17	12
S3_29	0.57	0.65	1568.054	17	-21.94	17	20
S3_30	0.53	0.64	2144.254	14	-24.54	17	4
S3_31	0.53	0.64	1792.139	9	-22.3	15	15
S3_32	0.52	0.64	1429.642	10	-20.51	17	13
S3_33	0.53	0.64	1890.874	10	-13.54	15	8
S3_34	0.55	0.65	1695.895	11	-22.17	15	20
S3_35	0.53	0.65	1740.649	8	-12.47	15	19

S3_36	0.54	0.64	1663.413	8	-16.66	15	23
S3_37	0.57	0.65	1824.085	14	-22.81	16	7
S3_38	0.57	0.65	1730.576	13	-18.79	15	4
S3_39	0.55	0.65	1581.347	10	-18.28	17	12
S3_40	0.54	0.65	1660.787	13	-13.69	15	23
S3_41	0.54	0.65	2050.673	9	-21.87	16	23
S3_42	0.53	0.64	1944.503	10	-22.39	15	17
S3_43	0.53	0.63	2004.581	11	-13.32	17	20
S3_44	0.53	0.65	2098.084	19	-13.35	17	16
S3_45	0.53	0.64	1592.321	9	-17.36	15	18
S3_46	0.58	0.65	2547.74	11	-20.49	16	8
S3_47	0.55	0.65	2075.042	19	-19.93	15	5
S3_48	0.55	0.64	1782.028	15	-24.12	15	9
S3_49	0.58	0.65	2292.741	10	-17.39	16	11
S3_50	0.53	0.64	1737.644	11	-12.57	15	19
S3_51	0.58	0.65	2452.06	13	-17.95	17	4
S3_52	0.58	0.65	1967.12	12	-13.93	17	7
S3_53	0.54	0.65	1594.367	14	-22.61	16	22
S3_54	0.54	0.65	2162.878	16	-17.37	17	24

S3_55	0.52	0.64	1630.13	10	-12.52	17	8
S3_56	0.53	0.64	1700.855	15	-15.27	17	5
S3_57	0.53	0.64	1455.658	10	-23.72	15	19
S3_58	0.53	0.65	1589.55	14	-14.92	16	12
S3_59	0.56	0.65	1916.114	17	-16.06	16	22
S3_60	0.54	0.65	1780.142	13	-24.53	15	21
S3_61	0.54	0.65	1674.459	17	-14.71	16	20
S3_62	0.53	0.64	1797.117	18	-21.31	17	5
S3_63	0.53	0.64	1624.347	17	-12.47	16	4
S3_64	0.55	0.64	1881.191	18	-20.08	17	8
S3_65	0.6	0.66	2050.174	14	-24.85	16	18
S3_66	0.52	0.64	2083.177	13	-15.81	17	18
S3_67	0.53	0.64	1887.649	10	-19.91	17	5
S3_68	0.53	0.64	1526.603	16	-23.14	15	9
S3_69	0.55	0.65	1597.863	15	-23.21	17	24
S3_70	0.55	0.65	1912.613	12	-13.38	16	13
S3_71	0.56	0.65	2068.513	18	-24.84	17	16
S3_72	0.57	0.66	1694.703	12	-20.96	16	17
S3_73	0.53	0.64	1517.825	19	-18.19	15	18

S3_74	0.54	0.65	2038.091	14	-23.08	16	17
S3_75	0.54	0.65	1792.975	9	-14.72	15	4
S3_76	0.52	0.64	1625.174	9	-19.03	15	18
S3_77	0.53	0.64	2079.439	17	-16.46	15	23
S3_78	0.53	0.64	1550.942	16	-24.92	17	8
S3_79	0.54	0.65	1702.658	15	-17.1	15	10
S3_80	0.53	0.64	1685.096	10	-23.99	15	20
S3_81	0.53	0.64	1979.595	10	-20.16	16	24
S3_82	0.53	0.64	1663.904	11	-18.96	17	15
S3_83	0.53	0.64	2028.658	17	-22.59	17	20
S3_84	0.53	0.64	1586.073	13	-16.55	15	5
S3_85	0.5	0.62	1714.102	19	-12.8	16	19
S3_86	0.54	0.64	2144.583	16	-22.4	15	13
S3_87	0.53	0.64	1576.106	14	-17.08	16	22
S3_88	0.58	0.65	1648.126	17	-21.81	15	4
S3_89	0.53	0.64	1453.153	19	-18.59	16	8
S3_90	0.56	0.65	1609.144	9	-14.56	16	19

A total of 90 independent AlphaFold3 predictions were performed and shown. ipTM, interface predicted TM-score (0–1, higher indicates higher confidence). pTM, predicted TM-score. BSA (Å²), buried surface area in square angstroms. For Hydrogen bond analysis, effective bonds denote the number of interfacial hydrogen bonds, and average energy represents the average interaction energy (in kcal/mol). mimic with IL6 denotes the number of interfacial hydrogen bonds that recapitulate the native binding interfaces of IL-6. For Hydrophobic pairing, effective bonds indicate the number of interfacial hydrophobic contacts.

Extended Table 12. Interface hotspot residues (Z-score > 2.00) identified by contact-based energetic scoring in the second round of evolution

Model	Residue	Chain	Score	Z-score
S1	TYR13	IL6	450	3.25
S1	ARG6	IL6	351	2.35
S1	LEU1	IL6	342	2.27
S1	TRP169	sIL6R	664.5	5.12
S1	TYR50	mimic	441	3.16
S1	TYR25	mimic	403	2.82
S2	TYR13	IL6	438.5	3.16
S2	ARG6	IL6	354	2.39
S2	LEU1	IL6	346	2.32
S2	TRP169	sIL6R	1880.5	8.21
S2	LEU172	sIL6R	841	3.3
S2	TYR50	mimic	431	3.09
S2	TYR25	mimic	382	2.64
S3	TYR13	IL6	437.5	3.14
S3	LEU1	IL6	358.5	2.43
S3	ARG6	IL6	349.5	2.34
S3	ARG118	sIL6R	260.5	4.43

S3	ARG115	sIL6R	205.5	3.35
S3	TYR50	mimic	440	3.16
S3	TYR25	mimic	387	2.68
S1	TYR13	IL6	450	3.25
S1	ARG6	IL6	351	2.35

List of predicted interface hotspot residues across the evolved models (S1, S2, and S3). Hotspots were defined as residues with a Z-score > 2.00 based on contact-based energetic scoring. Model, identifier of the evolved design model. Residue, amino acid type and sequence position of the identified hotspot. Chain, the binding interface context where the hotspot is located, where IL6 and sIL6R denote residues at the native IL-6 and sIL-6R binding interfaces, respectively, and mimic denotes residues on the binder interface that mimics the native binding mode. Score, the contact-based energetic score of the residue. Z-score, the standard score indicating the statistical significance of the energetic score relative to the background distribution.